



A Comparative Analysis of High-resolution Shock-capturing Schemes for Two-dimensional Magnetohydrodynamic Simulation of Flux Emergence in the Solar Atmosphere

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Abstract

This study presents a comparative analysis of high-resolution shock-capturing schemes in two-dimensional magnetohydrodynamic (MHD) simulations of magnetic flux emergence in the solar atmosphere. We evaluate four distinct reconstruction techniques based on recent improvements to the weighted essentially nonoscillatory (WENO) and targeted essentially nonoscillatory schemes. While these schemes have proven successful for the Euler equations of gas dynamics, their effectiveness in MHD simulations remains relatively unexplored. Our implementation combines the Harten–Lax–van Leer-discontinuities approximate Riemann solver for accurate flux computations, the generalized Lagrangian multiplier method for divergence control, and a third-order strong stability preserving the Runge–Kutta scheme for time integration. Numerical experiments reveal that these advanced schemes provide significant improvements in both accuracy and robustness in capturing complex MHD phenomena such as magnetosonic waves, MHD shocks, and magnetic buoyancy-driven instabilities. Among the tested methods, IMWENO-P proves to be the most physically consistent, effectively reproducing energy redistribution and compression patterns in line with theoretical predictions. These findings offer valuable insights into the strengths and limitations of each approach for simulating magnetic flux emergence dynamics.

Unified Astronomy Thesaurus concepts: [Solar magnetic fields \(1503\)](#); [Solar physics \(1476\)](#); [Magnetohydrodynamics \(1964\)](#); [Magnetohydrodynamical simulations \(1966\)](#)

1. Introduction

The solar atmosphere exhibits complex magnetohydrodynamic (MHD) behavior spanning multiple spatial and temporal scales, presenting significant challenges for numerical simulations (G. Tóth et al. 2012; M. Rempel 2016). The transport of magnetic flux from the convection zone into the solar atmosphere underlies numerous dynamic solar phenomena (T. Arber et al. 2007), with its emergence through the photosphere initiating a cascade of energetic processes (Y. Fan 2009; K. Shibata & T. Magara 2011). These processes, including solar flares and coronal mass ejections, can significantly impact Earth’s space environment, making their prediction crucial for space-weather forecasting (B. Schmieder & G. Aulanier 2012; K. Murawski et al. 2013; E. Priest 2014; Z. Liu et al. 2022).

The inherent complexity of magnetic flux emergence necessitates numerical methods capable of resolving multiscale interactions between plasma flows and magnetic fields while maintaining consistency with MHD conservation laws (J. E. Leake et al. 2013; M. C. M. Cheung & H. Isobe 2014; M. Rempel & M. C. M. Cheung 2014). This challenge is particularly acute in the solar atmosphere, where plasma parameters can vary by several orders of magnitude across relatively small spatial scales (T. Magara & D. Longcope 2003; J. Martínez-Sykora et al. 2017; V. Archontis & P. Syntelis 2019). Strong gradients and discontinuities that frequently develop during flux emergence events (P. Zhu et al. 2006;

T. Arber et al. 2007) present additional numerical challenges. Moreover, the low- β coronal environment demands careful treatment of the magnetic field evolution due to the dominance of magnetic forces over gas pressure (S. Inoue 2016; Y. Yang & F. Shen 2021; L. A. Tarr et al. 2024). These physical characteristics require numerical schemes that can maintain both accuracy and stability across vastly different plasma regimes, motivating our systematic evaluation of various high-resolution shock-capturing (HRSC) schemes for simulating solar atmospheric phenomena.

1.1. Background and Motivation

Historically, the modified Lax–Wendroff scheme (E. L. Rubin & S. Z. Burstein 1967) was among the earliest numerical approaches used for simulating magnetic flux emergence. Pioneering studies (e.g., K. Shibata et al. 1989a, 1989b, 1990; R. Matsumoto et al. 1990; T. Yokoyama & K. Shibata 1995; T. Magara 2001; S. Nozawa 2005; S. Toriumi & T. Yokoyama 2010) employed this method to simulate the emergence of magnetic flux and its interaction with the solar atmosphere. Their simulations revealed the evolution of emerging flux and its subsequent magnetic reconnection, providing valuable insights into the processes driving solar flares. Despite its contributions, the scheme has inherent limitations that restrict its applicability in flux emergence studies. Its reliance on a linearized, Taylor-expansion-based formulation without robust nonlinear limiters or characteristic-based reconstruction introduces explicit artificial viscosity (A. Harten 1983; A. Harten et al. 1987; H. C. Yee 1989; H. Yee 1997). The effectiveness of this artificial viscosity depends critically on a problem-dependent parameter (J. VonNeumann & R. D. Richtmyer 1950; C.-W. Shu 2009), making it challenging to maintain both



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stability and accuracy across different physical regimes typical in solar atmospheric simulations. Insufficient artificial viscosity leads to spurious oscillations near steep gradients and discontinuities (A. Harten & G. Zwas 1972; J. P. Boris & D. L. Book 1973; A. Lerat & R. Peyret 1974; R. F. Warming & B. Hyett 1974; M. Brio & C. C. Wu 1988), potentially generating unphysical states such as negative densities or pressures (J. P. Boris & D. L. Book 1973; A. Harten 1983). These numerical oscillations can persist regardless of grid resolution due to the scheme's dispersive nature (H. C. Yee 1989). Conversely, excessive artificial viscosity results in nonphysical diffusion that smooths out sharp features, obscuring the distinction between high-wavenumber physical processes and genuine discontinuities (H. C. Yee 1989; S. A. E. G. Falle 2002). Consequently, key structures like thin current sheets or steep gradient regions become either numerically unstable or overly diffused, hindering an accurate representation of these phenomena. The scheme's second-order accuracy further limits its capability to capture the intricate physical processes occurring at small spatial scales in the solar atmosphere. Additionally, the absence of robust magnetic divergence control mechanisms in earlier studies may introduce artificial forces parallel to the magnetic field lines due to accumulated divergence errors (J. U. Brackbill & D. C. Barnes 1980; D. Derigs et al. 2018). Such violations of the solenoidal constraint can lead to unphysical plasma acceleration along magnetic field lines and spurious energy generation (G. Tóth 2000; D. S. Balsara & J. Kim 2004). These limitations affect both precision and stability, particularly for problems requiring high accuracy and detailed resolution of small-scale features (J. W. Thomas 1995; R. J. LeVeque 2002; E. F. Toro 2009; D. Anderson et al. 2020).

To address these challenges, we integrate modern HRSC reconstruction schemes with the Harten–Lax–van Leer-discontinuities' (HLLD) approximate Riemann solver (T. Miyoshi & K. Kusano 2005), which effectively resolves contact discontinuities, Alfvén waves, and fast magnetosonic waves. We achieve higher-order accuracy through recent extensions of weighted essentially nonoscillatory (WENO) and targeted essentially nonoscillatory (TENO) schemes, providing accurate numerical flux calculations via the Riemann solver (C.-W. Shu 1998; D. S. Balsara & C.-W. Shu 2000). Additionally, we implement the generalized Lagrange multiplier (GLM) approach of A. Dedner et al. (2002) to maintain the solenoidal constraint, ensuring the physical consistency of the magnetic field evolution.

Comparative studies in hydrodynamics (e.g., E. F. Toro et al. 1994; D. S. Balsara & C.-W. Shu 2000; R. J. LeVeque 2002; G. Zhao et al. 2019; A. Hamzehloo et al. 2021; R. Li & W. Zhong 2022; L. Fu 2023) and MHD (e.g., J. M. Stone et al. 2008; M. Dumbser & E. F. Toro 2011; A. J. Christlieb et al. 2014; D. S. Balsara 2017; L. Fu & Q. Tang 2019) have established benchmarks for method evaluation. Modern frameworks such as Athena++ (K. G. Felker & J. M. Stone 2018), MPI-AMRVAC (C. Xia et al. 2018), CANS+ (Y. Matsumoto et al. 2019), PENCIL (The Pencil Code Collaboration et al. 2021), HOW-MHD (J. Seo & D. Ryu 2023), and MANCHA3D (M. Modestov et al. 2024) demonstrate that scheme selection significantly impacts simulation outcomes (J. H. Ferziger et al. 2002; H. Iijima 2016; K. G. Felker & J. M. Stone 2018; T. Baratashvili et al. 2023; R. Keppens et al. 2023). These findings highlight the need for comprehensive scheme evaluation and suggest that

optimal method choices may depend on problem-specific factors, including geometry and flow conditions.

1.2. Objectives

While recent advances in WENO and TENO schemes have demonstrated their effectiveness in hydrodynamic applications (e.g., H. Dong et al. 2019; L. Fu et al. 2019; L. Fu 2021; T. Liang et al. 2022; S. Takagi et al. 2022; A. Navas-Montilla & I. Echeverriabar 2023; T. Liang & L. Fu 2024; A. Navas-Montilla et al. 2024; X. Zhang et al. 2024), their performance in MHD simulations, particularly for astrophysical contexts, requires further investigation. Specifically, the interaction between these reconstruction methods, approximate Riemann solvers, and divergence control techniques needs systematic evaluation. A detailed comparative study can provide critical perspectives on the strengths and weaknesses of these methods, helping to identify the most suitable approaches for accurately simulating astrophysical MHD.

This study aims to provide a comprehensive analysis of HRSC schemes in modeling magnetic flux emergence, with three primary objectives. First, we evaluate the relative effectiveness of various reconstruction methods in capturing shocks, maintaining $\nabla \cdot \mathbf{B} = 0$, and resolving small-scale structures in the emerging flux. Second, we investigate the convergence properties and numerical stability of these schemes across multiple spatial resolutions, quantifying their ability to maintain physical fidelity at different scales. Third, we compare our results with the seminal works of K. Shibata et al. (1989a, 1989b) to establish the advantages of modern numerical methods in reproducing the complex dynamics of magnetic flux emergence.

The remainder of this paper is structured as follows. Section 2 presents the governing equations and model assumptions. Section 3 describes the numerical methodology, while Section 4 discusses the numerical experiments and their analysis. Finally, Section 5 concludes with key findings and future research directions.

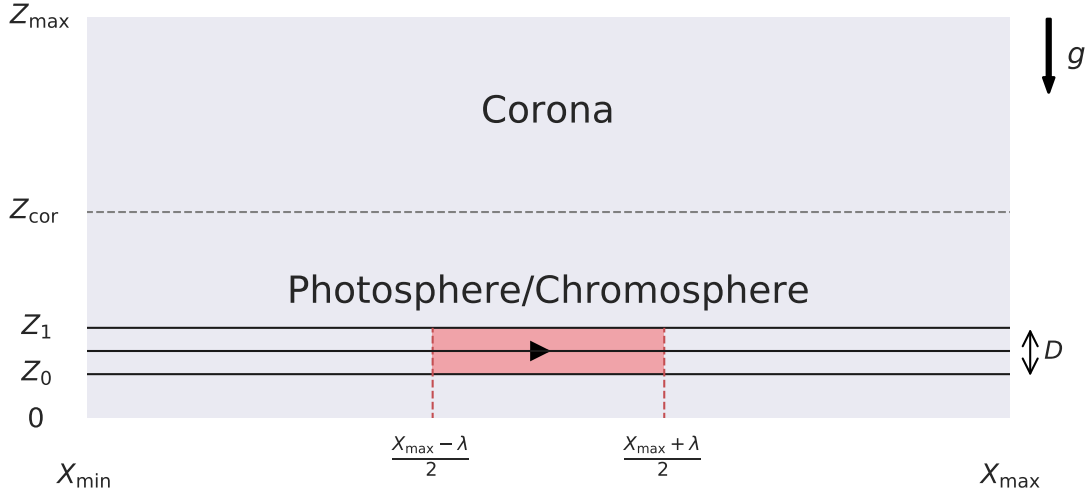
2. Theoretical Framework

This study focuses on the nonlinear evolution of the Parker (also known as the undular mode of the magnetic buoyancy) instability (E. N. Parker 1966) in a stratified solar atmosphere containing an isolated horizontal magnetic flux sheet (K. Shibata et al. 1989a, 1990). The system consists of two isothermal layers. As the instability develops, magnetic buoyancy drives the flux sheet upward, leading to its expansion into the corona through a self-similar evolution process (S. Toriumi & T. Yokoyama 2012). Previous studies (K. Shibata et al. 1989a, 1989b, 1990) have demonstrated that, in the lower atmosphere, this self-similar behavior is characterized by acceleration primarily driven by the magnetic pressure gradient force, which exceeds both gravitational and gas pressure gradient forces.

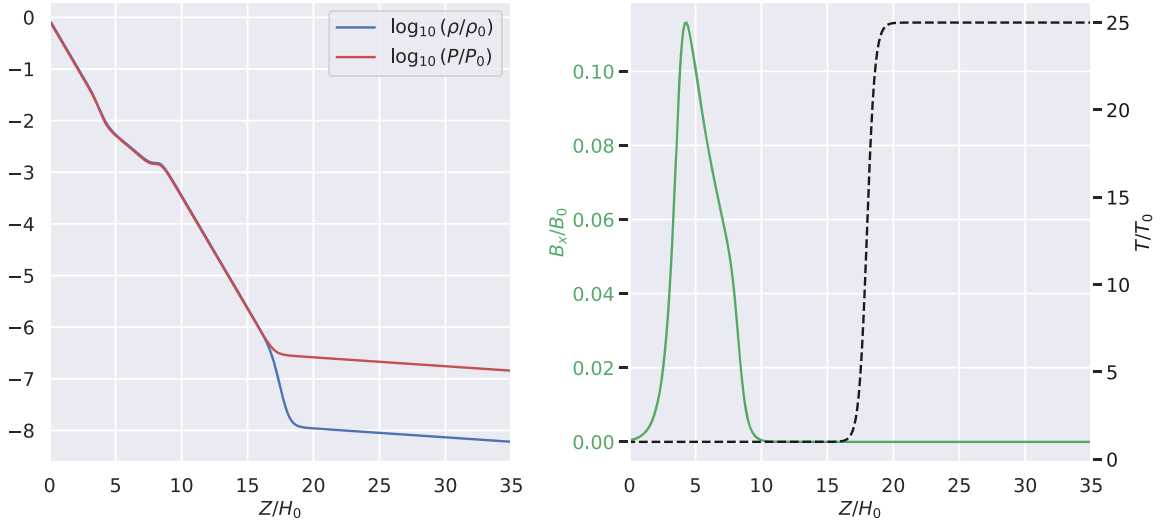
2.1. Initial and Boundary Conditions

Following K. Shibata et al. (1989a), we model the Parker instability of a single flux system and its subsequent emergence into a nonmagnetized corona (Figure 1(a)). The initial conditions are structured as follows:

1. *Two-temperature layered atmosphere.* The initial configuration consists of a magnetostatic equilibrium with a cold isothermal plasma layer containing a horizontal



(a) Schematic of the initial magnetic flux configuration



(b) Initial vertical profiles

Figure 1. Initial setup. (a) Schematic showing a magnetic flux sheet positioned between heights z_0 and z_1 , with initial velocity perturbation (red patch) and temperature transition from chromospheric (T_{ch}) to coronal (T_{cor}) values near z_{cor} . (b) Vertical profiles showing density (solid blue) and pressure (dashed red) in logarithmic scale (left), and temperature (dashed black) and magnetic field strength (solid green) profiles (right), illustrating the stratified atmosphere and localized magnetic flux sheet.

magnetic flux sheet ($z_0 < z < z_1$), overlaid by a hot isothermal, nonmagnetized plasma layer. The temperature distribution follows

$$T(z) = T_{\text{ch}} + \frac{(T_{\text{cor}} - T_{\text{ch}})}{2} \left[\tanh\left(\frac{z - z_{\text{cor}}}{w_{\text{tr}}}\right) + 1 \right], \quad (1)$$

where T_{ch} and T_{cor} are the chromospheric and coronal temperatures respectively, z_{cor} is the coronal base height, and w_{tr} is the temperature scale height in the transition region. We set $T_{\text{cor}} = 25T_0$, $T_{\text{ch}} = T_0$, $z_{\text{cor}} = 18H_0$, and $w_{\text{tr}} = 0.6H_0$, with H_0 being the photospheric/chromospheric pressure scale height.

2. *Magnetic flux sheet.* The initial magnetic field, parallel to the x -axis and localized within the cold layer, has a

strength distribution of

$$B(z) = \left[\frac{8\pi p(z)}{\beta(z)} \right]^{1/2}, \quad \beta(z) = \frac{\beta_*}{f(z)}, \quad (2)$$

where

$$f(z) = \frac{1}{4} \left[1 + \tanh\left(\frac{z - z_0}{w_0}\right) \right] \left[1 - \tanh\left(\frac{z - z_1}{w_1}\right) \right]. \quad (3)$$

Here, β_* is the plasma beta (ratio of gas to magnetic pressure) at the flux sheet center. The magnetic flux sheet is bounded vertically by $z_0 = 4H_0$ (lower boundary) and $z_1 = z_0 + D$ (upper boundary), where $D = 4H_0$ defines the sheet thickness. The function $f(z)$ ensures transitions at the sheet boundaries using hyperbolic tangent

Table 1
Units of Physical Quantities in cgs Units

Quantity	Symbol	Definition	Units
Length	H_0	$\frac{k_B T_0}{m g_0}$	[cm]
Time	τ_0	$\frac{H_0}{C_s}$	[s]
Velocity	C_s	$\left(\frac{\gamma k_B T_0}{m}\right)^{1/2}$	[cm s ⁻¹]
Density	ρ_0	...	[g cm ⁻³]
Pressure	p_0	$\rho_0 C_s^2$	[dyne cm ⁻²]
Temperature	T_0	$\frac{m C_s^2}{\gamma k_B}$	[K]
Magnetic field	B_0	$(\rho_0 C_s^2)^{1/2}$	[G]
Gravitational acceleration	g_0	$\frac{C_s^2}{\gamma H_0}$	[cm s ⁻²]
Energy	E_0	$\rho_0 H_0^3 C_s^2$	[erg]

Note. Here, k_B and m denote the Boltzmann constant and mean molecular mass, respectively.

functions, with transition widths $w_0 = w_1 = 0.5H_0$ controlling the steepness at both boundaries.

3. *Equilibrium state.* The initial magnetostatic equilibrium balances magnetic pressure gradient, gas pressure gradient, and gravity through

$$\frac{dp_{\text{total}}}{dz} = -\rho g, \quad p_{\text{total}}(z) = p(z) + \frac{B_x^2(z)}{8\pi}, \quad (4)$$

where g is the gravitational acceleration.

4. *Perturbations.* To trigger the Parker instability, we introduce horizontal velocity perturbations within the magnetic flux sheet ($z_0 < z < z_1$) over the central domain region ($X_{\text{max}}/2 - \lambda/2 < x < X_{\text{max}}/2 + \lambda/2$), where $\lambda = 20H_0$ is the perturbation wavelength, and $X_{\text{max}} = 80H_0$ is the horizontal domain size. The perturbation takes the form

$$V_x = f(z) A C_s \sin \left[\frac{2\pi(x - X_{\text{max}}/2)}{\lambda} \right], \quad (5)$$

with amplitude $A = 0.05$ relative to the sound speed C_s . Here, $f(z)$ is the same transition function defined earlier.

For the boundary conditions, we apply periodic conditions horizontally, symmetric conditions at the bottom ($z = 0$), and free conditions with an absorbing layer at the top boundary ($z = Z_{\text{max}} = 35H_0$) to prevent artificial wave reflections (M. Machida & R. Matsumoto 2003).

Figure 1(b) shows the initial distributions of temperature, density, pressure, and magnetic field strength. All physical quantities are normalized to their photospheric/chromospheric values: length by the pressure scale height (H_0), velocity by the adiabatic sound speed (C_s), time by the sound-crossing time (H_0/C_s), and density by the reference density (ρ_0). Table 1 summarizes these normalization units and their definitions in cgs units.

2.2. The GLM-MHD Model

We use a conservative discretization of compressible ideal MHD equations augmented with the hyperbolic–parabolic divergence cleaning technique proposed by A. Dedner et al. (2002). This approach, known as the GLM-MHD method, introduces a scalar field ψ that couples to the magnetic field

evolution to maintain the divergence-free condition $\nabla \cdot \mathbf{B} = 0$. The resulting system of GLM-MHD equations can be written in conservative form:

$$\frac{\partial \mathbf{U}}{\partial t} + \sum_{l=x,y,z} \frac{\partial \mathbf{F}_l}{\partial l} = \mathbf{S}, \quad (6)$$

where a state vector of conserved variables $\mathbf{U}$ and the corresponding vector of fluxes $\mathbf{F}_l$ are defined as

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho V_x \\ \rho V_y \\ \rho V_z \\ B_x \\ B_y \\ B_z \\ e \\ \psi \end{bmatrix}, \quad \mathbf{F}_l = \begin{bmatrix} \rho V_l \\ \rho V_x V_l - B_l B_x + \delta_{lx} \left(p + \frac{\|\mathbf{B}\|^2}{2} \right) \\ \rho V_y V_l - B_l B_y + \delta_{ly} \left(p + \frac{\|\mathbf{B}\|^2}{2} \right) \\ \rho V_z V_l - B_l B_z + \delta_{lz} \left(p + \frac{\|\mathbf{B}\|^2}{2} \right) \\ B_x V_l - V_x B_l + \delta_{lx} \psi \\ B_y V_l - V_y B_l + \delta_{ly} \psi \\ B_z V_l - V_z B_l + \delta_{lz} \psi \\ \left(e + p + \frac{\|\mathbf{B}\|^2}{2} \right) V_l - B_l (\mathbf{V} \cdot \mathbf{B}) \\ c_h^2 B_l \end{bmatrix},$$

$$\mathbf{S} = \begin{bmatrix} 0 \\ \rho g_x \\ \rho g_y \\ \rho g_z \\ 0 \\ 0 \\ 0 \\ \rho (\mathbf{V} \cdot \mathbf{g}) \\ -\frac{c_h^2}{c_p^2} \psi \end{bmatrix}. \quad (7)$$

Here, ρ is the mass density, $\mathbf{V}$ is the velocity, $\mathbf{B}$ is the magnetic field, and e is the total energy density. The thermal pressure p is given by

$$p = (\gamma - 1) \left(e - \frac{1}{2} \rho \|\mathbf{V}\|^2 - \frac{1}{2} \|\mathbf{B}\|^2 \right), \quad (8)$$

where γ is the adiabatic index, taken to be 1.05 in this paper. This reduced value of γ is chosen as it leads to a higher growth rate of the undular mode in the magnetic buoyancy instability (E. N. Parker 1966; K. Shibata et al. 1990) and mimics quasi-isothermal heating with limited energy injection in the coronal temperature (J. Guo et al. 2024). These equations are expressed using Heaviside–Lorentz units, which redefine the electric charge by a factor of $1/4\pi$, thereby eliminating the 4π factors in Maxwell’s equations (A. Beresnyak 2019).

The eigenstructure of the GLM-MHD system reveals nine characteristic waves (A. Dedner et al. 2002; A. Mignone et al. 2010). The eigenvalues of the flux Jacobian $\frac{\partial \mathbf{F}_l}{\partial \mathbf{U}}$ in the x -directions are

$$\lambda^{1,9} = \mp c_h, \quad \lambda^{2,8} = V_x \mp c_f, \quad \lambda^{3,7} = V_x \mp c_a, \\ \lambda^{4,6} = V_x \mp c_s, \quad \lambda^5 = V_x, \quad (9)$$

with characteristic speeds

$$c_{f,s} = \left[\frac{\gamma p + \|\mathbf{B}\|^2 \pm \sqrt{(\gamma p + \|\mathbf{B}\|^2)^2 - 4\gamma p B_x^2}}{2\rho} \right]^{1/2},$$

$$c_a = \frac{|B_x|}{\sqrt{\rho}}. \quad (10)$$

Beyond the conventional MHD modes (c_f , c_s , c_a) representing fast magnetosonic, slow magnetosonic, and Alfvén speeds, the additional modes ($\lambda^{1,9} = \mp c_h$) govern the propagation of divergence errors through the coupling of B_x and ψ (A. Mignone & P. Tzeferacos 2010). These modes propagate at the maximum allowable signal speed:

$$c_h = \max(|V_x| + c_{f,x}, |V_y| + c_{f,y}, |V_z| + c_{f,z}), \quad (11)$$

where ($c_{f,x}$, $c_{f,y}$, $c_{f,z}$) are the fast magnetosonic speeds in each direction. The cleaning speed c_h is typically determined from the Courant stability condition in Equation (16) as it represents the fastest characteristic wave (Y. Matsumoto et al. 2019). The effectiveness of divergence cleaning depends on proper parameter selection. The key is achieving critical damping of the divergence waves by properly balancing diffusive and advective timescales (T. S. Tricco et al. 2016). Initially, A. Dedner et al. (2002) proposed a fixed ratio $c_p^2/c_h = 0.18$, but this prescription proved inadequate due to its dimensional nature. A. Mignone & P. Tzeferacos (2010) and A. Mignone et al. (2010) later introduced a dimensionless parameter α_p , defined as the ratio of diffusive (Δt_d) to advective (Δt_a) timescales

$$\alpha_p = \frac{\Delta t_d}{\Delta t_a} = \frac{\Delta h^2/c_p^2}{\Delta h/c_h} = \Delta h \frac{c_h}{c_p^2}, \quad (12)$$

where $\Delta h = \min(\Delta x, \Delta z)$ is the smallest mesh spacing. Optimal divergence cleaning is achieved for $\alpha_p \in [0, 1]$, with the specific value depending on the problem and numerical scheme (A. Mignone & P. Tzeferacos 2010; A. Mignone et al. 2010). An analysis of magnetic field divergence errors for different α_p values is presented in Section 4.2. For further details on the implementation of the GLM-MHD model, see A. Dedner et al. (2002), A. Mignone & P. Tzeferacos (2010), A. Mignone et al. (2010), J. Vides et al. (2013), T. S. Tricco et al. (2016), and Y. Matsumoto et al. (2019).

3. Numerical Methods

We employ a two-dimensional finite volume (FV) discretization of Equation (6) on a uniform Cartesian mesh with cell sizes $\Delta x \times \Delta z$, where indices (i, k) denote the cell centers at coordinates (x_i, z_k). Integrating the conservation equations over discrete cells and applying the divergence theorem yield (R. J. LeVeque 2002)

$$\frac{d\tilde{U}_{(i,k)}}{dt} = - \frac{(\hat{F}_x(i+\frac{1}{2},k) - \hat{F}_x(i-\frac{1}{2},k))}{\Delta x} - \frac{(\hat{F}_z(i,k+\frac{1}{2}) - \hat{F}_z(i,k-\frac{1}{2}))}{\Delta z} + \tilde{S}_{(i,k)} := \mathcal{L}(\tilde{U}), \quad (13)$$

where $\tilde{U}$ and $\tilde{S}$ are the cell-averaged values of the state vector U and source term S , respectively. $\mathcal{L}(\tilde{U})$ defines a spatial discretization operator that can be coupled with various time

integration schemes (D. Drikakis & W. Rider 2005). The numerical fluxes $\hat{F}$ represent spatial averages of the physical fluxes F over cell interfaces. These fluxes are evaluated using the HLLD approximate Riemann solver, which resolves the complex wave structure inherent in MHD flows while maintaining computational efficiency (T. Miyoshi & K. Kusano 2005; X. Guo et al. 2016; see Section 3.2).

The semidiscrete formulation in Equation (13), often referred to as the method of lines, separates spatial and temporal discretizations. This decoupling enables independent selection of spatial discretization schemes and time integration methods, potentially achieving higher-order accuracy in both domains (A. Suresh & H. Huynh 1997; G. Tóth 2000). While this formulation is spatially exact, derived from the integral conservation law, approximations enter through two channels: the evaluation of numerical fluxes at cell interfaces and the chosen time-stepping procedure (J. M. Stone et al. 2008). In particular, the ultimate spatial accuracy depends on how the interface states are reconstructed.

The subsequent sections describe the time-stepping process, numerical flux function, and reconstruction methods.

3.1. Time-stepping Process

For the time integration of our semidiscrete system, we utilize the third-order strong stability-preserving Runge–Kutta (SSP-RK3) scheme. This method, originally developed by C.-W. Shu & S. Osher (1988) and further analyzed by S. Gottlieb & C.-W. Shu (1998), is particularly well suited for problems involving discontinuities or sharp gradients, as it preserves the strong stability properties of the underlying spatial discretization. The SSP-RK3 scheme is implemented as follows:

$$\begin{aligned} \mathbf{u}^{(0)} &= \tilde{U}^n, \\ \mathbf{u}^{(1)} &= \mathbf{u}^{(0)} + \Delta t^n \mathcal{L}(\mathbf{u}^{(0)}), \\ \mathbf{u}^{(2)} &= \frac{3}{4} \mathbf{u}^{(0)} + \frac{1}{4} [\mathbf{u}^{(1)} + \Delta t^n \mathcal{L}(\mathbf{u}^{(1)})], \\ \mathbf{u}^{(3)} &= \frac{1}{3} \mathbf{u}^{(0)} + \frac{2}{3} [\mathbf{u}^{(2)} + \Delta t^n \mathcal{L}(\mathbf{u}^{(2)})], \\ \tilde{U}^{n+1} &= \mathbf{u}^{(3)}, \end{aligned} \quad (14)$$

where $\tilde{U}^n$ represents the solution at time step n , and Δt^n is the time step size. The time step size Δt^n is restricted by the Courant–Friedrichs–Levy (CFL) condition:

$$\Delta t^n = \lambda_{\text{cfl}} \min \left[\frac{\Delta x}{\max_i (|\lambda|_i)}, \frac{\Delta z}{\max_k (|\lambda|_k)} \right], \quad (15)$$

where $\lambda_l^{2,8}$ ($l = i, k$) are the fast magnetosonic wave speeds along each dimension, and λ_{cfl} is the Courant number. Alternatively, using the hyperbolic cleaning speed c_h from Equation (11), the CFL condition can be expressed as

$$\Delta t^n = \frac{\lambda_{\text{cfl}}}{c_h} \min(\Delta x, \Delta z). \quad (16)$$

Throughout this study, we fix $\lambda_{\text{cfl}} = 0.2^3$ for all schemes and resolutions to ensure stable and consistent comparisons.

³ S. Tang (2024) reported that certain TENO schemes (TENO-A and TENO-LAD; L. Fu et al. 2018; J. Peng et al. 2021) encounter stability issues at $\lambda_{\text{cfl}} \geq 0.3$ in two-dimensional configurations.

3.2. Numerical Flux Function

To compute the numerical fluxes at cell interfaces, we employ the HLLD approximate Riemann solver (T. Miyoshi & K. Kusano 2005), which provides reliable approximations for MHD systems by resolving intermediate wave structures such as Alfvén, contact, and slow discontinuities (A. Mignone et al. 2009). The solver operates on reconstructed variables in a characteristic-wise manner to enhance accuracy and stability.

In our two-dimensional implementation, the numerical flux at the cell interface $(i + \frac{1}{2}, k)$ in the x -direction is computed as

$$\hat{F}_x(i+\frac{1}{2},k) = \text{HLLD}\left({}^L\hat{\mathbf{V}}_{(i+\frac{1}{2},k)}, {}^R\hat{\mathbf{V}}_{(i+\frac{1}{2},k)}\right), \quad (17)$$

where $\text{HLLD}(\dots, \dots)$ denotes the flux computed from the left $({}^L\hat{\mathbf{V}}_{(i+\frac{1}{2},k)})$ and right $({}^R\hat{\mathbf{V}}_{(i+\frac{1}{2},k)})$ states of the interpolated primitive variables. These states are obtained through

$${}^{L,R}\hat{\mathbf{V}}_{(i+\frac{1}{2},k)} = \mathbf{R}_{x(i,k)} {}^{L,R}\hat{\mathbf{W}}_{(i+\frac{1}{2},k)}, \quad (18)$$

where $\mathbf{R}_{x(i,k)}$ is the right eigenvector matrix of the Jacobian $\frac{\partial \mathbf{F}_x}{\partial \mathbf{U}}$ evaluated at cell (i, k) . The characteristic variables ${}^{L,R}\hat{\mathbf{W}}_{(i+\frac{1}{2},k)}$ are reconstructed using a high-order reconstruction scheme $\mathcal{R}$ (see Section 3.3):

$${}^{L,R}\hat{\mathbf{W}}_{(i+\frac{1}{2},k)} = \mathcal{R}(\tilde{\mathbf{W}}_{(i+s,k)}), \quad (19)$$

where $\tilde{\mathbf{W}}_{(i+s,k)}$ are the local characteristic variables at $(i + s, k)$, with the stencil width $s \in [-r, r]$ for a $(2r + 1)$ -th-order scheme. These characteristic variables are computed as

$$\tilde{\mathbf{W}}_{(i+s,k)} = \mathbf{L}_{x(i,k)} \tilde{\mathbf{V}}_{(i+s,k)}, \quad (20)$$

where $\mathbf{L}_{x(i,k)}$ denotes the left eigenvector matrix, and $\tilde{\mathbf{V}}_{(i+s,k)}$ represents the cell-averaged primitive variables within the stencil.

This characteristic-wise reconstruction approach enhances the scheme's accuracy and stability (J. Qiu & C.-W. Shu 2002), particularly near discontinuities (O. Zanotti & M. Dumbser 2016). The effectiveness of the HLLD solver is intrinsically linked to the quality of spatial reconstructions, necessitating careful evaluation of various reconstruction schemes and their interaction with the flux solver (J. M. Stone et al. 2008).

3.3. Reconstruction Method

For the reconstruction of left and right interface states in Equation (19), we employ multiple HRSC schemes: the classical WENO-JS scheme (G.-S. Jiang & C.-W. Shu 1996) and several recent developments including WENO-Z+M (X. Luo & S.-p. Wu 2021), WENO-NZ (S. Han & M. Li 2023), IMWENO-P (W. Zhong et al. 2023), and TENO-Local Adaptive Dissipation (TENOLAD; J. Peng et al. 2021). All schemes discussed in this study are based on fifth-order accurate formulations.

For clarity of presentation, we focus on the reconstruction procedure for the left state in the x -direction, denoting it as ${}^L\hat{\mathbf{W}}_{(i+\frac{1}{2})}$ by omitting the k index. The right state ${}^R\hat{\mathbf{W}}_{(i+\frac{1}{2})}$ follows an analogous reconstruction process.

3.3.1. Weighted Essentially Nonoscillatory Schemes

WENO schemes are designed to achieve high-order accuracy in smooth regions while effectively capturing discontinuities without introducing spurious oscillations (X.-D. Liu et al. 1994; G.-S. Jiang & C.-W. Shu 1996; D. S. Balsara & C.-W. Shu 2000). The key idea is to construct a weighted combination of lower-order polynomials from different substencils to obtain a high-order approximation of the solution (Y.-T. Zhang & C.-W. Shu 2016). Through this approach, the contribution of stencils containing discontinuities is minimized by assigning them lower weights (G.-S. Jiang & C.-W. Shu 1996; for a comprehensive review and historical development of the WENO schemes, see C.-W. Shu 2009).

For the fifth-order WENO scheme, the left-biased interface value ${}^L\hat{\mathbf{W}}_{(i+\frac{1}{2})}$ is expressed as

$${}^L\hat{\mathbf{W}}_{(i+\frac{1}{2})} = \sum_{k=0}^2 \omega_k {}^L\hat{\mathbf{W}}_{(i+\frac{1}{2})}^{(k)}, \quad (21)$$

where ω_k is nonlinear weights satisfying $\sum_{k=0}^2 \omega_k = 1$ and $\omega_k \geq 0$. The terms ${}^L\hat{\mathbf{W}}_{(i+\frac{1}{2})}^{(k)}$ represent third-order Lagrange polynomials reconstructed over three-point substencils S_k , defined as

$$\begin{aligned} S_0 &= \{I_{i-2}, I_{i-1}, I_i\}, \\ S_1 &= \{I_{i-1}, I_i, I_{i+1}\}, \\ S_2 &= \{I_i, I_{i+1}, I_{i+2}\}, \end{aligned}$$

with I_i denoting the cell centered at position i . These third-order polynomials are given by

$$\begin{aligned} {}^L\hat{\mathbf{W}}_{(i+\frac{1}{2})}^{(0)} &= \frac{1}{3}\tilde{\mathbf{W}}_{i-2} - \frac{7}{6}\tilde{\mathbf{W}}_{i-1} + \frac{11}{6}\tilde{\mathbf{W}}_i, \\ {}^L\hat{\mathbf{W}}_{(i+\frac{1}{2})}^{(1)} &= -\frac{1}{6}\tilde{\mathbf{W}}_{i-1} + \frac{5}{6}\tilde{\mathbf{W}}_i + \frac{1}{3}\tilde{\mathbf{W}}_{i+1}, \\ {}^L\hat{\mathbf{W}}_{(i+\frac{1}{2})}^{(2)} &= \frac{1}{3}\tilde{\mathbf{W}}_i + \frac{5}{6}\tilde{\mathbf{W}}_{i+1} - \frac{1}{6}\tilde{\mathbf{W}}_{i+2}, \end{aligned} \quad (22)$$

where $\tilde{\mathbf{W}}_i$ represents the characteristic variables at the cell center i .

Various WENO variants have been developed that differ primarily in their strategies for computing the nonlinear weights ω_k , which we examine in the following sections.

WENO-JS scheme. The fifth-order WENO-JS scheme, introduced by G.-S. Jiang & C.-W. Shu (1996), defines its nonlinear weights ω_k as

$$\omega_k^{JS} = \frac{\alpha_k^{JS}}{\sum_{l=0}^2 \alpha_l^{JS}}, \quad \alpha_k^{JS} = d_k \left(\frac{1}{\epsilon + \beta_k} \right)^p, \quad (23)$$

where d_k are ideal weights ($d_0 = 0.1$, $d_1 = 0.6$, $d_2 = 0.3$) that yield the fifth-order central upwind scheme (S. Han & M. Li 2023), $p \geq 1$ controls the weights of less-smooth substencils, and ϵ is a positive sensitivity parameter to prevent division by zero. Notably, M. Castro et al. (2011) demonstrated that ϵ functions as a cutoff for the smoothness measure. In this study, we employ $p=2$ and $\epsilon = 10^{-40}$. The local smoothness indicators (LSIs) β_k , which measure the roughness of the solution within each substencil, are

defined as

$$\begin{aligned}\beta_0 &= \frac{13}{12}(\tilde{W}_{i-2} - 2\tilde{W}_{i-1} + \tilde{W}_i)^2 \\ &\quad + \frac{1}{4}(\tilde{W}_{i-2} - 4\tilde{W}_{i-1} + 3\tilde{W}_i)^2, \\ \beta_1 &= \frac{13}{12}(\tilde{W}_{i-1} - 2\tilde{W}_i + \tilde{W}_{i+1})^2 \\ &\quad + \frac{1}{4}(\tilde{W}_{i-1} - \tilde{W}_{i+1})^2, \\ \beta_2 &= \frac{13}{12}(\tilde{W}_i - 2\tilde{W}_{i+1} + \tilde{W}_{i+2})^2 \\ &\quad + \frac{1}{4}(3\tilde{W}_i - 4\tilde{W}_{i+1} + \tilde{W}_{i+2})^2.\end{aligned}\quad (24)$$

While the WENO-JS scheme effectively captures discontinuities, it exhibits two significant limitations (A. K. Henrick et al. 2005; R. Borges et al. 2008): (i) degraded accuracy at critical points, and (ii) excessive numerical diffusion⁴ in nominally smooth flow regions due to its nonlinear weighting mechanism, which artificially dampens physically realistic oscillations associated with instability development (A. Wang et al. 2021a). These limitations have motivated the development of numerous improved variants.

WENO-NZ scheme. The WENO-NZ scheme (S. Han & M. Li 2023) introduces an improved global smoothness indicator (GSI) to address limitations of the fifth-order WENO-Z scheme (R. Borges et al. 2008), particularly its inability to balance low diffusion with fifth-order convergence at critical points. The nonlinear weights are given by

$$\begin{aligned}\omega_k^{\text{NZ}} &= \frac{\alpha_k^{\text{NZ}}}{\sum_{l=0}^2 \alpha_l^{\text{NZ}}}, \quad \alpha_k^{\text{NZ}} = d_k \left(1 + \left(\frac{\tau_5^{\text{NZ}}}{\epsilon + \beta_k} \right)^p \right), \\ p &= 1 \text{ or } 2,\end{aligned}\quad (25)$$

where τ_5^{NZ} represents the novel GSI, defined as

$$\begin{aligned}\tau_5^{\text{NZ}} &= \frac{|\beta_{00} - 3\beta_{01} + 3\beta_{12} - \beta_{22}|}{6} \quad \text{with} \\ \begin{cases} \beta_{00} &= (\tilde{W}_{i-2} - \tilde{W}_{i-1})^2, \\ \beta_{01} &= (\tilde{W}_{i-1} - \tilde{W}_i)^2, \\ \beta_{12} &= (\tilde{W}_i - \tilde{W}_{i+1})^2, \\ \beta_{22} &= (\tilde{W}_{i+1} - \tilde{W}_{i+2})^2. \end{cases}\end{aligned}\quad (26)$$

A key feature of WENO-NZ is its decomposition of three three-point substencils $\{S_0, S_1, S_2\}$ into four two-point substencils $\{S_{00}, S_{01}, S_{12}, S_{22}\}$, which minimizes the influence of discontinuities on interpolation. This approach provides two main advantages: (i) larger weights for less-smooth substencils, resulting in reduced numerical diffusion, and (ii) nonlinear weights that more closely approximate the ideal weights of the central upwind scheme. However, sharing the same theoretical convergence order as WENO-Z, the WENO-NZ scheme may still exhibit suboptimal convergence at higher-order critical points. As suggested by S. Han & M. Li (2023), we implement the scheme with $p = 2$.

⁴ While commonly referred to as “dissipation” in many works, the smoothing effect of WENO schemes is more accurately viewed as “diffusion,” as it redistributes, rather than removes, conserved quantities (A. Wang et al. 2021a).

WENO-Z+M scheme. The WENO-Z+ scheme (F. Acker et al. 2016) enhanced the WENO-Z scheme (R. Borges et al. 2008) by increasing weights of less-smooth substencils, to improve small-scale structure resolution. However, despite its improvements, WENO-Z+ exhibited negative diffusion in certain spectral ranges and stability issues. To address these limitations, X. Luo & S.-p. Wu (2021) introduced the WENO-Z+M scheme, which modifies the weight structure to eliminate negative diffusion while preserving high spectral properties. The nonlinear weights are computed as

$$\begin{aligned}\omega_k^{\text{ZPM}} &= \frac{\alpha_k^{\text{ZPM}}}{\sum_{l=0}^2 \alpha_l^{\text{ZPM}}}, \quad \alpha_k^{\text{ZPM}} = d_k [1 + (\xi_k)^p + \lambda^{\text{ZPM}} \eta_k^{\text{ZPM}}], \\ p &= 2,\end{aligned}\quad (27)$$

where ξ_k and τ_5^Z are inherited from the fifth-order WENO-Z weights. The modified terms η_k^{ZPM} and λ^{ZPM} are defined as

$$\eta_k^{\text{ZPM}} = (\eta_k^{\text{ZP}})^{\frac{1}{2}} \quad \text{with} \quad \eta_k^{\text{ZP}} = \frac{1}{\xi_k}, \quad (28)$$

$$\lambda^{\text{ZPM}} = \lambda^{\text{ZP}} \cdot z \quad \text{with} \quad z = \frac{1 + \xi_{\min}^2}{\sum_{k=0}^2 d_k (1 + \xi_k^2)}, \quad (29)$$

$$\begin{aligned}\xi_{\min} &= \min(\xi_0, \xi_1, \xi_2), \quad \xi_k = \frac{\epsilon + \tau_5^Z}{\epsilon + \beta_k}, \\ \tau_5^Z &= |\beta_2 - \beta_0|.\end{aligned}\quad (30)$$

While the original η_k^{ZP} enhanced resolution by increasing less-smooth substencil weights, its excessive influence led to negative diffusion. The modified η_k^{ZPM} moderates this effect, effectively eliminating the negative diffusion observed in WENO-Z+ (F. Acker et al. 2016). The adaptive parameter λ^{ZPM} dynamically regulates the nonlinear weight contributions across different flow regimes: in smooth regions, it allows weights to closely approximate their ideal values, preserving high resolution for small-scale structures; near discontinuities, it reduces less-smooth substencil contributions to prevent numerical oscillations while maintaining stability.

IMWENO-P scheme. The IMWENO-P scheme (W. Zhong et al. 2023) builds upon previous WENO schemes, including WENO-NS (Y. Ha et al. 2013), WENO-P (C. H. Kim et al. 2016), and MWENO-P schemes (S. Rathen & G. N. Raju 2018). Its key advancement lies in the introduction of novel LSIs and GSIs that eliminate user-tunable parameters while preserving symmetry properties. These symmetry properties can enhance WENO schemes by improving stability, reducing diffusion, and sharpening resolution (W. S. Don et al. 2018; R. Abedian 2020; X. Liu & N. R. Morgan 2020). The LSIs are defined as

$$\begin{aligned}\beta_0^{\text{IMP}} &= 4\Delta x (\tilde{W}_{i-2} - 3\tilde{W}_{i-1} + 2\tilde{W}_i)^2 \\ &\quad + (\tilde{W}_{i-2} - 2\tilde{W}_{i-1} + \tilde{W}_i)^2, \\ \beta_1^{\text{IMP}} &= \Delta x (\tilde{W}_{i-1} - \tilde{W}_{i+1})^2 + (\tilde{W}_{i-1} - 2\tilde{W}_i + \tilde{W}_{i+1})^2, \\ \beta_2^{\text{IMP}} &= 4\Delta x (2\tilde{W}_i - 3\tilde{W}_{i+1} + \tilde{W}_{i+2})^2 \\ &\quad + (\tilde{W}_i - 2\tilde{W}_{i+1} + \tilde{W}_{i+2})^2\end{aligned}\quad (31)$$

where Δx denotes the cell spacing in the x -direction. The GSI is formulated as

$$\zeta^{\text{IMP}} = (\tilde{W}_{i-2} - 4\tilde{W}_{i-1} + 6\tilde{W}_i - 4\tilde{W}_{i+1} + \tilde{W}_{i+2})^2. \quad (32)$$

Using these symmetry-preserving indicators, the nonlinear weights are computed as

$$\omega_k^{\text{IMP}} = \frac{\alpha_k^{\text{IMP}}}{\sum_{l=0}^2 \alpha_l^{\text{IMP}}}, \quad \alpha_k^{\text{IMP}} = d_k \left(1 + \left(\frac{\zeta^{\text{IMP}}}{\epsilon + \beta_k^{\text{IMP}}} \right)^p \right), \quad (33)$$

where p is typically set to 1. Through a rigorous theoretical analysis, W. Zhong et al. (2023) proved that ω_k^{IMP} satisfies the sufficient condition⁵ $\omega_k - d_k = \mathcal{O}(\Delta x^3)$ for optimal fifth-order convergence (A. K. Henrick et al. 2005), while achieving even higher accuracy:

$$\omega_k^{\text{IMP}} - d_k = \begin{cases} \mathcal{O}(\Delta x^6) & \text{for noncritical points,} \\ \mathcal{O}(\Delta x^5) & \text{for first-order critical points,} \\ \mathcal{O}(\Delta x^3) & \text{for second-order critical points.} \end{cases} \quad (34)$$

This enhanced convergence to the ideal weights d_k in smooth regions ensures superior preservation of high-order accuracy compared to other WENO schemes. Comprehensive numerical experiments by W. Zhong et al. (2023) showed that IMWENO-P achieves several significant improvements over its predecessors: optimal convergence orders near critical points, reduced numerical diffusion near discontinuities, and enhanced resolution of fine-scale flow structures.

3.3.2. Targeted Essentially Nonoscillatory Scheme

The family of TENO schemes has been developed to address key limitations of WENO schemes (L. Fu et al. 2016, 2017, 2018, 2019; L. Fu & Q. Tang 2019; L. Fu 2021, 2023; J. Peng et al. 2021; S. Tang 2024). TENO schemes combine an innovative ENO-like stencil selection with a scale separation technique (X. Hu & N. A. Adams 2011), significantly enhancing discontinuity detection while minimizing numerical diffusion in smooth regions. This approach allows TENO schemes to recover the spectral properties of corresponding linear schemes, particularly in low to intermediate wavenumber ranges. The nonlinear weights in TENO schemes are computed as

$$\omega_k^{\text{TENO}} = \frac{\alpha_k^{\text{TENO}}}{\sum_{l=0}^2 \alpha_l^{\text{TENO}}}, \quad \alpha_k^{\text{TENO}} = \delta_k d_k, \quad (35)$$

where δ_k is a binary cutoff function

$$\delta_k = \begin{cases} 0, & \chi_k < C_T, \\ 1, & \text{otherwise.} \end{cases} \quad (36)$$

⁵ According to A. K. Henrick et al. (2005), the sufficient condition ensures that the nonlinear weights ω_k converge to the ideal weights d_k within $\mathcal{O}(\Delta x^3)$ accuracy, which is essential for maintaining optimal fifth-order convergence at critical points in smooth regions.

The normalized smoothness measure χ_k is defined as

$$\chi_k = \frac{\gamma_k}{\sum_{l=0}^2 \gamma_l}, \quad \gamma_k = \left(C + \frac{\tau_5^Z}{\epsilon + \beta_k} \right)^p, \quad (37)$$

where the scale-separation mechanism is controlled by parameters C and p (X. Hu & N. A. Adams 2011), typically set to $C=1$ and $p=6$. Following J. Peng et al. (2021), we employ τ_5^Z as the GSI. The scheme's diffusion properties are governed by the cutoff value C_T , typically set to 10^{-6} , although this value requires careful tuning for specific applications. While smaller C_T values reduce numerical diffusion, they may introduce numerical oscillations (S. Tang 2024).

TENO-LAD scheme. The TENO-LAD scheme (J. Peng et al. 2021) addresses a key limitation of conventional TENO schemes: the manual tuning requirement of C_T to balance the shock-capturing capability with numerical dissipation. This is accomplished through a local adaptive method that dynamically adjusts C_T based on the smoothness of the reconstruction stencil. Building upon the TENO-Adaptive (TENO-A) scheme (L. Fu et al. 2018), TENO-LAD effectively controls numerical oscillations near discontinuities while maintaining low diffusion in smooth regions. This adaptive approach improves robustness across diverse flow scenarios without requiring case-specific parameter tuning, making it particularly suitable for high-fidelity compressible flow simulations (J. Peng et al. 2021). The adaptive C_T is determined by

$$\begin{cases} C_T = 10^{-m}, \\ m = B_l + \lfloor \theta(B_u - B_l) \rfloor \end{cases} \quad (38)$$

with

$$\theta = \frac{1}{1 + \left(\max_k \frac{\tilde{\chi}_k}{H} \right)}, \quad \tilde{\chi}_k = \frac{\tau_5^Z}{\epsilon + \beta_k}, \quad \begin{cases} B_l = 4, \\ B_u = 10, \\ H = 10. \end{cases} \quad (39)$$

Here, $\lfloor \dots \rfloor$ denotes the floor function, and θ represents the Runge function, which serves as both a shock sensor (J. Peng & Y. Shen 2015) and a switcher (J. Peng et al. 2019) in an adaptive manner. The parameter H defines the intensity threshold for discontinuities, while B_l and B_u establish the lower and upper bounds for the adaptive C_T , respectively. The computation of nonlinear weights follows the conventional TENO formulation as specified in Equations (35)–(37).

4. Numerical Results and Discussion

In this section, we present numerical experiments designed to evaluate the performance of various HRSC schemes in simulating magnetic flux emergence. We assess their effectiveness in resolving key physical features while maintaining numerical stability and accuracy.

4.1. Nonlinear Evolution of Parker Instability

We analyze the nonlinear evolution across multiple resolutions ranging from 300^2 to 2400^2 (Figure 2). These figures present snapshots at $t = 51\tau_0$ for four advanced HRSC schemes, illustrating magnetic field lines, velocity vectors, and density distributions. Magnetic field lines are visualized as isocontours of A_y in the $y=0$ plane, where A_y is the y -component of the magnetic

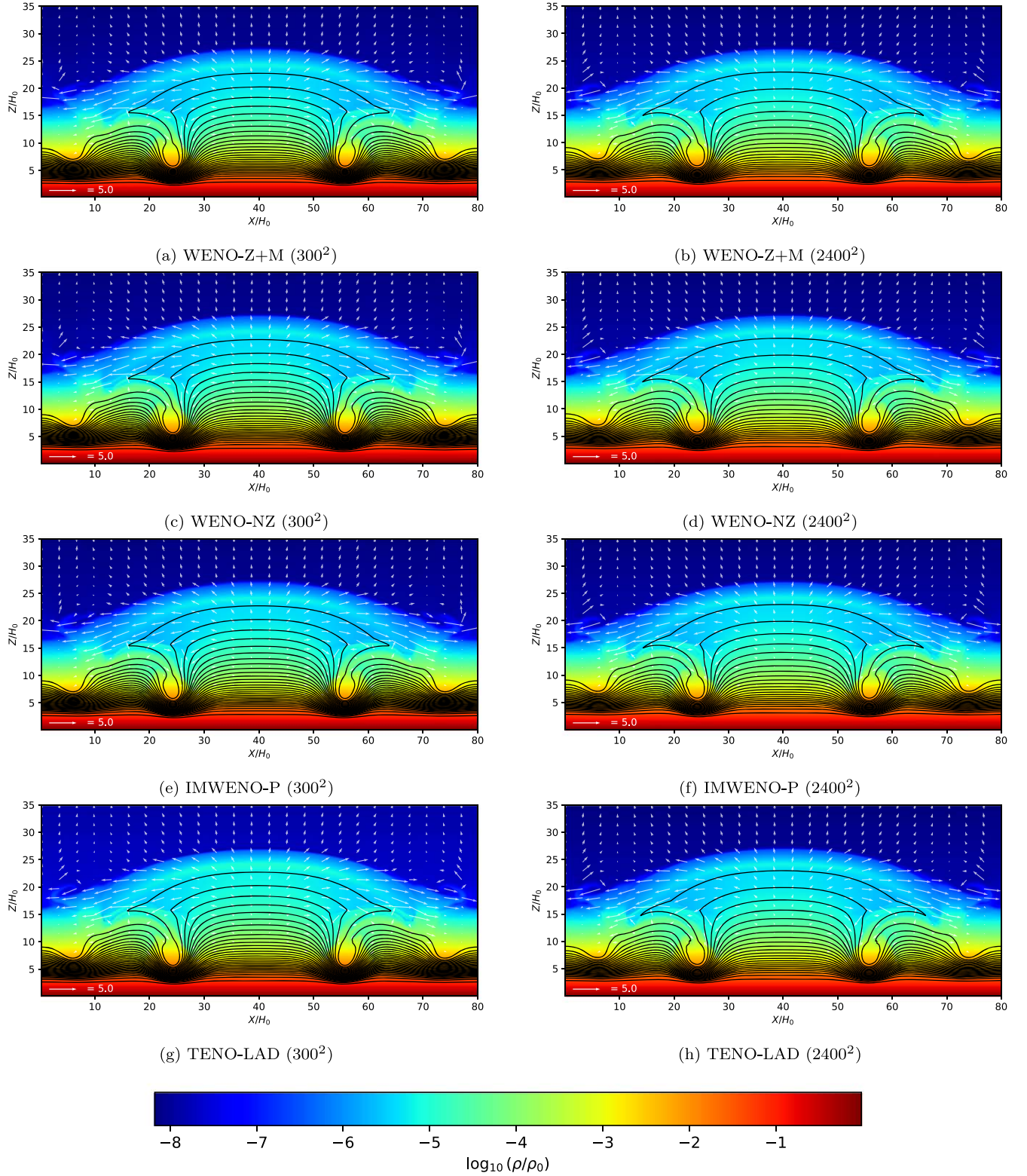


Figure 2. Two-dimensional visualization of magnetic field lines (black curves), velocity vectors (white arrows), and density distribution (color) at $t = 51\tau_0$ using different numerical schemes. Left column shows results at 300^2 resolution; right column at 2400^2 resolution. While all schemes capture the essential features at both resolutions, the refined mesh reveals more detailed flow structures and sharper discontinuities.

vector potential defined by $\mathbf{B} = \nabla \times \mathbf{A}$. In this slice, the in-plane magnetic field components satisfy $B_x = -\partial A_y / \partial z$ and $B_z = \partial A_y / \partial x$. Thus, contours of constant A_y delineate the magnetic field lines in the x - z plane. A complete temporal evolution sequence is shown in Appendix A (Figure 14).

As the instability evolves, all schemes consistently capture the following key physical processes:

1. the buoyant rise of magnetic flux accompanied by downward gas flow along the forming loop, leading to matter accumulation in magnetic pockets at the loop base,

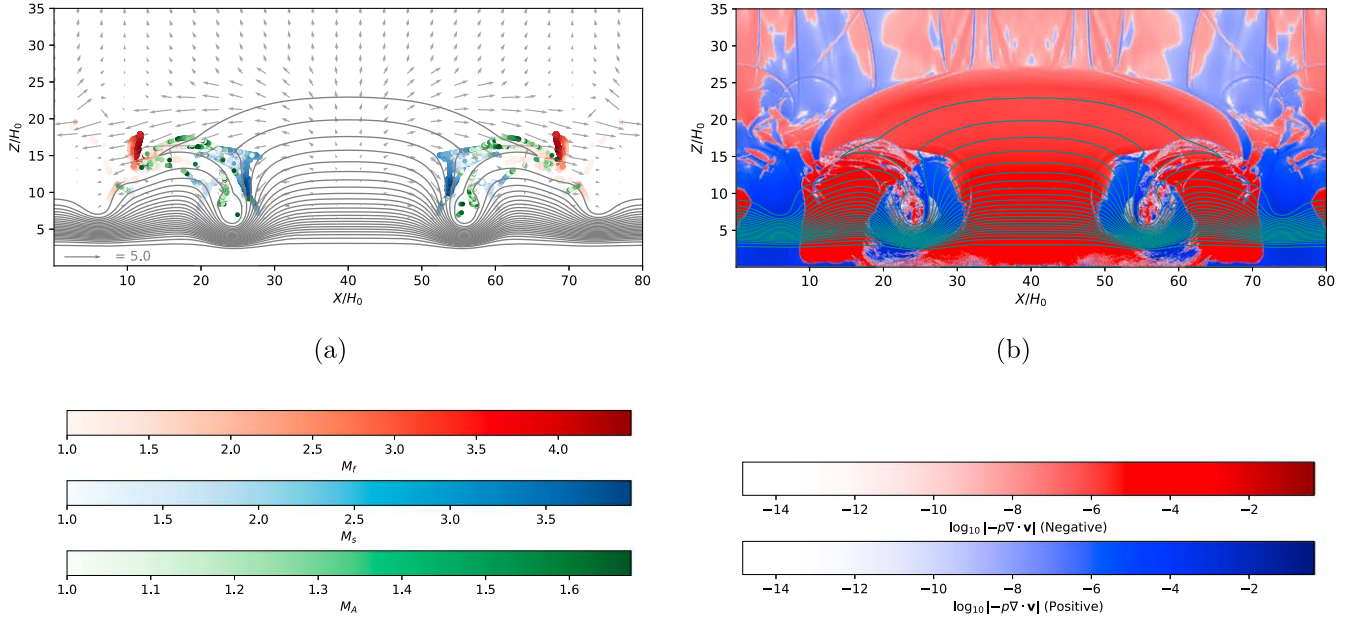


Figure 3. MHD shock analysis using IMWENO-P scheme at 2400^2 and $t = 51\tau_0$. (a) Potential shock locations. The background displays magnetic field lines (gray contours) and velocity vectors (gray arrows). Shocks are categorized as fast (red), slow (blue), and intermediate (green), with darker shades indicating higher Mach numbers (M_f , M_s , M_A). Three color bars show the normalized Mach numbers for each shock type, where $M > 1$ signifies the presence of a shock. (b) Pressure work $(-p \nabla \cdot \mathbf{V})$ distribution with magnetic field lines (dark cyan contours), where red and blue regions indicate expansion (negative) and compression (positive) in logarithmic scale.

- the nearly self-similar expansion of magnetic loops accelerated by the magnetic pressure gradient force,
- the formation of fast and intermediate MHD shocks along the expanding loop sides, where the lateral downflow speeds ($V \simeq 4C_s - 5C_s$) exceed both the local sound and Alfvén speeds ($V_A \simeq 2C_s - 3C_s$).

While these evolution patterns share qualitative features with previous studies on two-dimensional magnetic flux emergence (K. Shibata et al. 1989a, 1989b, 1990), we identify previously underemphasized vertical downflows along magnetic field lines near loop footpoints (Figure 3(a)). The previous studies focused on lateral downflows and their associated shock formation during loop expansion (K. Shibata et al. 1989a, 1989b), which we also observe. However, our results reveal field-aligned vertical downflows near loop footpoints, characterized by supersonic yet sub-Alfvénic downflow speeds ($V_z \approx 2C_s - 3C_s$) relative to the ambient Alfvén speed ($V_A \approx 5C_s - 7C_s$). The supersonic yet sub-Alfvénic nature creates favorable environments for slow MHD shock formation, where the component of the magnetic field parallel to the shock plane ($B_{\parallel}$) decreases in magnitude across the shock front (H. De Sterck et al. 1998). Notably, the presence of slow-mode waves near footpoints aligns with recent solar observations, where slow-mode MHD waves have been frequently observed in regions of strong magnetic field concentration (P. Kumar et al. 2013; T. Wang et al. 2021b; A. Rawat & G. R. Gupta 2023). Additionally, in areas of strong magnetic shear and complex plasma flows (i.e., turbulent plasmas), particularly near current sheets, intermediate shocks may form (K. Shibata et al. 1989b; B. Snow et al. 2021; S. Wang et al. 2024). In these environments, the $B_{\parallel}$ undergoes not only a change in magnitude but often a reversal of direction.

We further analyze the correlation between pressure work $(-p \nabla \cdot \mathbf{V})$ and MHD shock formation. As illustrated in Figure 3, we observe distinct relationships between compression patterns and shock types. Fast and slow shocks predominantly develop in regions of strong compression, evidenced by significant positive values of pressure work. In contrast, intermediate shocks

emerge in regions of complex flow convergence, where opposing plasma streams generate intricate pressure work patterns. These regions of flow interaction exhibit rich MHD dynamics, suggesting that the interplay between magnetic and fluid forces may facilitate intermediate shock formation. However, it is important to note that our two-dimensional configuration requires careful interpretation as it may enhance the apparent abundance of intermediate shocks due to the simplified treatment of out-of-plane magnetic field dynamics and the sensitivity of shock detection criteria.

Among the schemes considered, we rely on IMWENO-P’s shock detection results as the most physically consistent representation of the underlying flow dynamics. While other schemes produce negative values of the integrated compressional work $\int_{\Omega} (-p \nabla \cdot \mathbf{V}) dA$, IMWENO-P maintains physically plausible values near zero or slightly positive, indicating it does not artificially induce global expansion or compression (Figure 4). This numerical robustness stems from IMWENO-P’s refined weighting and smoothness criteria, which enable a selective diffusion application without overdiffusing compressive features, resulting in more balanced treatment of the velocity field and associated compressive waves. Consequently, IMWENO-P’s pressure work evolution aligns with the expected behavior of the quasi-isothermal system, ensuring that observed thermal energy increases originate primarily from the gravitational energy release rather than unphysical compressional heating (detailed analysis in Section 4.3).

To quantitatively analyze the self-similar evolution, we examine four key parameters: rising velocity, local Alfvén speed, horizontal magnetic field, and density. The evolution of these parameters follows the theoretical expansion law (K. Shibata et al. 1989a):

$$V_z/C_s = a_1 \Delta z, \quad (40)$$

$$V_A/C_s = a_2 \Delta z, \quad (41)$$

$$\rho/\rho_0 \propto (\Delta z)^{-4}, \quad (42)$$

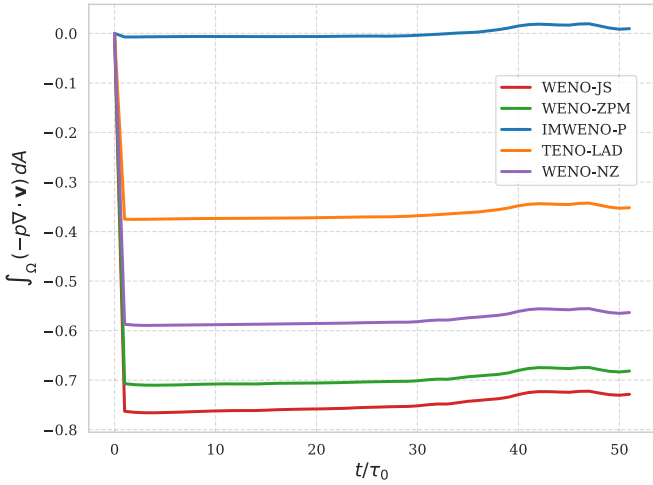


Figure 4. Time evolution of integrated pressure work ($\int_{\Omega} (-p \nabla \cdot \mathbf{V}) dA$) using different numerical schemes at 2400^2 resolution and $t = 51\tau_0$, where Ω denotes the computational domain, and $dA = \Delta x \Delta z$ is the cell area.

$$B_x/B_0 \propto (\Delta z)^{-1}, \quad (43)$$

where $a_1 = 0.062$, $a_2 = 0.3$ are constants, and $\Delta z = (z - z_0)/H_0$ is the dimensionless height measured from the base of the magnetic flux sheet (see Table 1 in K. Shibata et al. 1989a).

At the coarse resolution (Figure 5), all schemes demonstrate consistent nonlinear growth patterns governed by the expansion law. However, the TENO-LAD exhibits Gibbs phenomena near strong discontinuities, such as regions of low density with supersonic flows (Figures 5(d) and (p)). This artifact arises from the binary stencil cutoff mechanism in Equation (36), where a stencil is fully deactivated ($\delta_k = 0$) if the smoothness measure χ_k falls below the cutoff threshold C_T . Although the C_T dynamically increases near discontinuities to enhance diffusion, the abrupt deactivation of stencils due to marginal χ_k values often fails to stabilize underresolved regions with steep gradients. This instability can be further exacerbated by order-degeneration near higher-order critical points.⁶ Consequently, the intended contributions of stencils are disrupted, leading to an insufficient suppression of high-frequency oscillations and the observed Gibbs phenomena. Therefore, the TENO-LAD appears particularly susceptible to numerical instabilities in near-vacuum regions with high-Mach-number flows (L. Fu & K. Griffin 2021). This observation aligns with previous findings that emphasize the inherent limitations of the TENO-LAD scheme (see S. Tang 2024, for details).

At finer resolutions (Figures 6(a) and (d)), these oscillations diminish as finer meshes enable the LSIs (β_k) and GSIs (τ_k^Z) to more accurately assess local flow features. At the finest resolution (2400^2), all schemes converge to nearly identical profiles. To avoid redundancy, we present only the results from the TENO-LAD scheme at this resolution (Figure 6). Results for other schemes at different resolutions are provided in Appendix B.

We further note that the local Alfvén speed increases due to the evacuation of plasma density by the downflows along the loop. This leads to a substantial decrease in the local β as

⁶ While the TENO-LAD achieves optimal accuracy at first-order critical points, its accuracy degrades at second- and third-order critical points. This order-degeneration occurs because the LSIs (β_k) converge to similar values, causing the nonlinear weights (ω_k) to substantially deviate from the ideal weights (d_k) (S. Tang 2024).

shown in Figure 6(e) ($\beta \simeq (V_A/C_s)^2$ and $C_s \simeq \text{const}$ due to $\gamma = 1.05$). The magnetic pressure thus becomes increasingly dominant over the gas pressure ($\beta \ll 1$), driving continued expansion of the magnetic loop (S. Nozawa 2005).

A comprehensive description of the expansion law can be found in K. Shibata et al. (1989a, 1989b, 1990).

4.2. Divergence-free Analysis

To quantitatively analyze how $\nabla \cdot \mathbf{B}$ evolves, we introduce a modified divergence error metric. The conventional local metric for measuring the magnetic field divergence error is expressed as

$$\epsilon_{\text{local}} = \frac{\Delta L \|\nabla \cdot \mathbf{B}\|}{\|\mathbf{B}\|}, \quad (44)$$

where ΔL is the characteristic length scale. However, this local measure becomes numerically unstable in regions where $\|\mathbf{B}\|$ falls below the mesh resolution scale. Given our initial magnetic field configuration (Figure 1), which includes nonmagnetized regions where $\|\mathbf{B}\|$ approaches extremely small values ($\sim 10^{-18}$), we adopt a more robust global error metric:

$$\epsilon_{\text{global}} = \frac{\int_{\Omega} \Delta L \|\nabla \cdot \mathbf{B}\| dA}{\int_{\Omega} \|\mathbf{B}\| dA}, \quad (45)$$

where Ω denotes the entire computational domain, and $dA = \Delta x \Delta z$ is the discrete cell area. Figure 7 shows the temporal evolution of ϵ_{global} for all schemes across different mesh resolutions. Our analysis reveals an inverse relationship between numerical diffusion and magnetic field divergence control. The WENO-JS scheme exhibits consistently smaller divergence errors due to its higher numerical diffusion. In contrast, the TENO-LAD scheme, while better preserving sharp magnetic features, shows larger divergence errors due to its reduced numerical diffusion. Figure 8 illustrates the spatial distribution of $\Delta L \|\nabla \cdot \mathbf{B}\|$ ($\Delta L = \sqrt{\Delta x \Delta z}$), where TENO-LAD shows elevated local errors near current sheets, particularly in regions of high magnetic field curvature. This behavior is further elucidated by examining the integrated current density:

$$\int_{\Omega} \|\mathbf{J}\| dA \quad \text{with} \quad \mathbf{J} = \nabla \times \mathbf{B}. \quad (46)$$

Figure 9 shows that WENO-JS produces lower current density magnitudes due to its larger smoothing effect on the field structure, while TENO-LAD preserves stronger current features by maintaining sharp field gradients. This trade-off between numerical diffusion and divergence control reflects fundamental challenges in discretizing steep magnetic field gradients, particularly in regions of rapid directional changes.

At coarser resolutions (300^2 and 600^2), WENO-family schemes exhibit pronounced spurious oscillations during the initial phase (Figures 9(a) and (b)). These numerical artifacts originate from the schemes' treatment of the vertical velocity component V_z , which emerges from the initial horizontal velocity perturbation specified in Equation (5). Although V_z varies smoothly, it remains several orders of magnitude smaller than other flow variables. In regions of extremely low-amplitude variations, the LSIs (β_k) can become numerically unreliable due to round-off errors (E. Johnsen 2008; E. Johnsen & F. Ham 2012; W. S. Don et al. 2018). This numerical

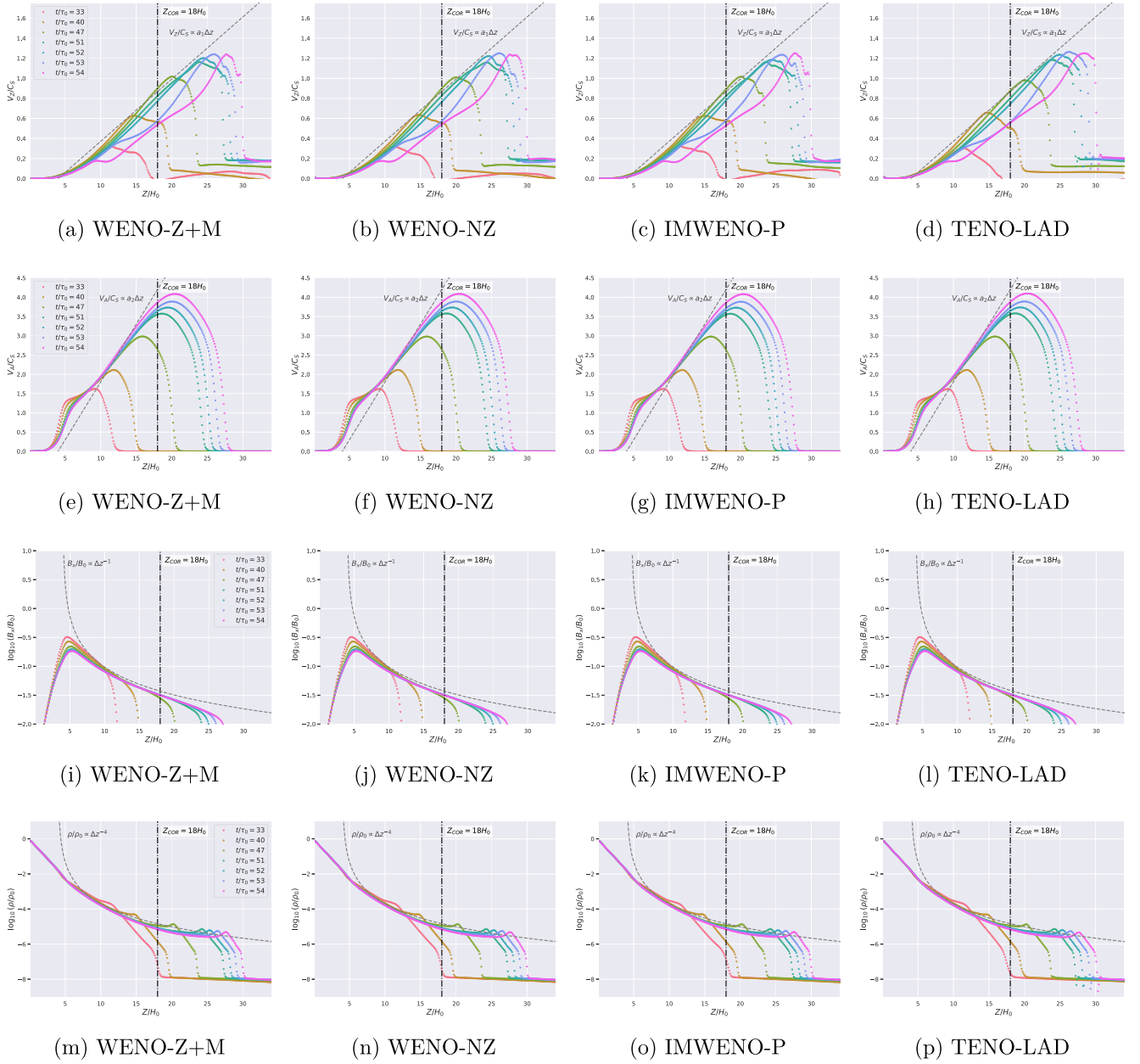


Figure 5. Comparison of key physical quantities at 300^2 resolution using different numerical schemes. First row (a)–(d): rising velocity (V_z/C_s). Second row (e)–(h): Alfvén speed (V_A/C_s). Third row (i)–(l): horizontal magnetic field (B_x/B_0). Fourth row (m)–(p): density (ρ/ρ_0). All quantities are evaluated at $x/H_0 = (X_{\max}/2)/H_0$, corresponding to the midpoint of the rising loop. Gray and black dashed lines represent theoretical densities and coronal height respectively. Our results exhibit sharper transitions near the contact surface compared to K. Shibata et al. (1989a; see their Figures 7 and 12), demonstrating improved resolution of the magnetic loop structure.

instability can be further amplified by the combination of a small parameter value ($\epsilon = 10^{-40}$) and coarse mesh resolution, as the scheme’s accuracy strongly depends on both factors (A. K. Henrick et al. 2005). As a result, the nonlinear weights erroneously interpret these minuscule gradients as discontinuities, assigning inappropriate stencil weights and often producing ill-conditioned or even negative weights⁷ (J. Shi et al. 2002; J. Qiu & C.-W. Shu 2002; C.-W. Shu 2009; W. S. Don et al. 2018). This chain of errors creates artificial discontinuities in V_z , which propagate into the

magnetic field computation and are amplified by the curl operation, resulting in significant oscillations in the current density field. The TENO-LAD scheme mitigates these issues by employing a binary activation step and then reverting to its standard weights in Equations (36) and (35). This approach effectively decouples the decision to retain or discard a stencil from the very small amplitude features, ensuring that genuinely smooth, low-amplitude variations are not misidentified as discontinuities.⁸ By using fixed optimal weights for activated stencils, TENO-LAD prevents the amplification of numerical

⁷ With $\epsilon = 10^{-40}$, small negative random β_k values would diminish the influence of ϵ in even- p schemes (as in WENO-JS, WENO-Z+M, and WENO-NZ), introducing unnecessary random substencil bias in underresolved or discontinuous stencils. For an odd- p scheme (as in IMWENO-P), such values may destabilize the solution by occasionally producing negative weights (W. S. Don et al. 2018).

⁸ As described in Section 3.3.2, the TENO weighting strategy inherently serves as a discontinuity sensor, classifying local flow scales as smooth or nonsmooth. This approach differs from WENO-family schemes that preferentially select smoother stencils through convex combinations, thereby avoiding the occurrence of negative weights (L. Fu et al. 2016; L. Fu 2023).

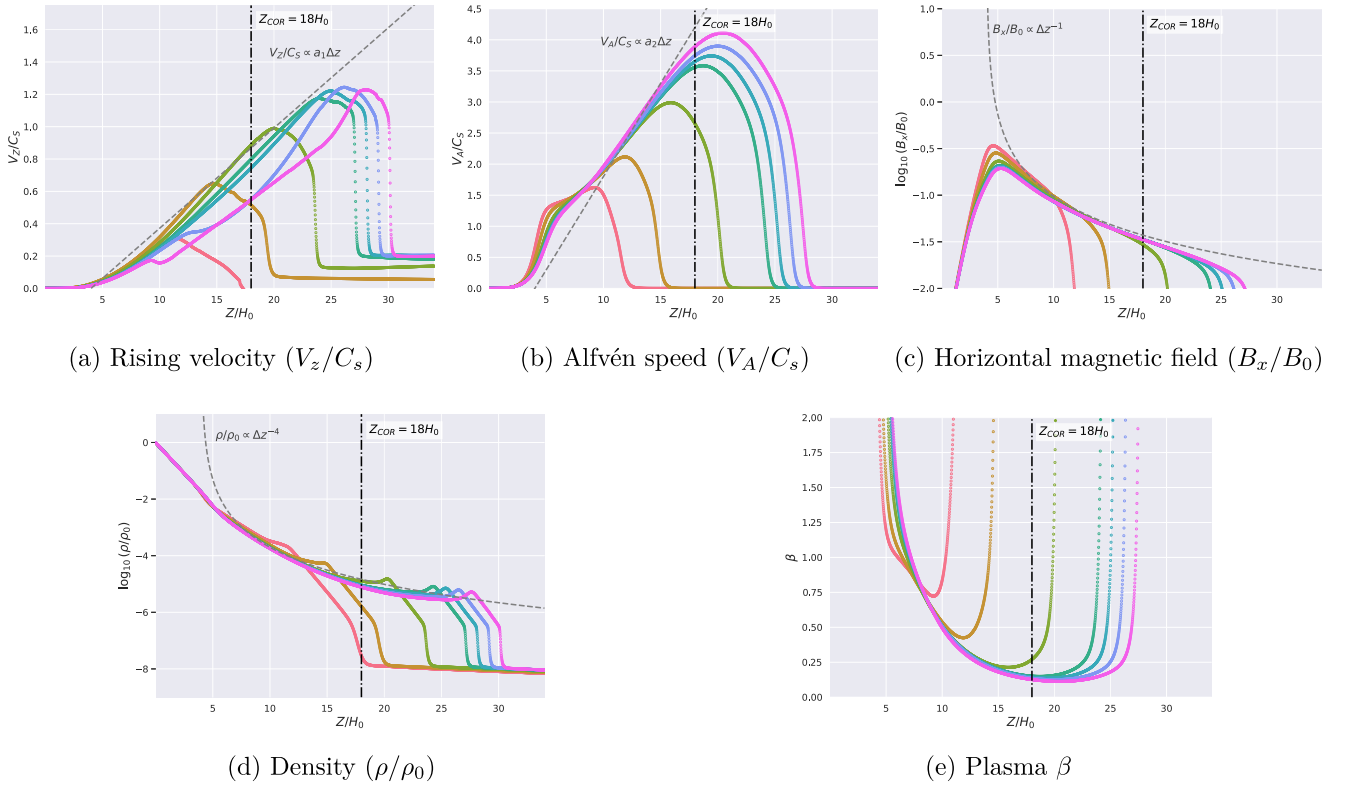


Figure 6. Key physical quantities using the TENO-LAD scheme at 2400^2 resolution: (a) rising velocity, (b) Alfvén speed, (c) horizontal magnetic field, (d) density, and (e) plasma beta. Measurements are taken at the same evaluation location as in Figure 5, i.e., at $x/H_0 = (X_{\max}/2)/H_0$.

noise through weight fluctuations, effectively reducing instabilities arising from round-off-induced anomalies.

In the context of solar flux emergence simulations, this analysis reveals a fundamental trade-off in numerical MHD: schemes that better maintain $\nabla \cdot \mathbf{B} = 0$ through higher diffusion tend to smooth out physically important sharp magnetic structures. Conversely, low-diffusion schemes better capture these sharp features but at the cost of larger divergence errors. This balance between divergence control and preservation of sharp magnetic structures remains a key consideration in numerical modeling of solar magnetic phenomena. To optimize divergence control in the GLM-MHD framework, we analyze the effect of the damping parameter α_p in Equation (12) (A. Mignone & P. Tzeferacos 2010; A. Mignone et al. 2010).

The effectiveness of divergence control is quantified using the L1 norm of $\nabla \cdot \mathbf{B}$, which measures the deviation from the ideal zero-divergence constraint. Mathematically, the L1 norm is defined as

$$\epsilon_1 = \int_{\Omega} \|\nabla \cdot \mathbf{B}\| dA. \quad (47)$$

Our parametric study (Figure 10) reveals scheme-dependent characteristics: WENO-family schemes consistently achieve optimal divergence control at $\alpha_p \approx 0.2$ across all resolutions. In contrast, TENO-LAD shows resolution-dependent behavior, with optimal performance at $\alpha_p \approx 0.2$ for coarser resolutions (300^2 and 600^2) but requiring $\alpha_p \approx 0.3$ at finer resolution (1200^2). Based on these results, we adopt $\alpha_p = 0.2$ for all WENO-family schemes, and $\alpha_p = 0.3$ for TENO-LAD at finer resolutions (1200^2 and 2400^2).

4.3. Energetics

We now analyze the temporal evolution of different forms of energy to understand how the numerical schemes influence energy conservation and conversion processes. According to I. B. Bernstein et al. (1958), when an adiabatic equation of state is employed, the total energy of the system should be conserved even in the presence of shock waves. Consequently, any changes in kinetic energy (E_k), magnetic energy (E_m), and gravitational energy (E_g) should be balanced by corresponding changes in thermal energy (E_{th}) through compressional and shock heating processes (K. Shibata et al. 1989b). This relationship can be expressed as

$$\Delta E_k + \Delta E_m + \Delta E_g = -\Delta E_{\text{th}}, \quad (48)$$

where $\Delta E = E(t) - E(0)$ denotes the change in each energy component from its initial value. Figure 11 illustrates the temporal evolution of magnetic, kinetic, gravitational, and thermal energies across different mesh resolutions. Notably, the impact of different numerical schemes on the energy conversion processes is evident. At coarser resolutions (300^2 and 600^2), WENO-family schemes exhibit excessive thermal energy generation, indicating significant numerical dissipation that artificially converts kinetic and magnetic energies into thermal energy. Conversely, TENO-LAD shows relatively lower thermal energy generation due to its low-diffusion design, although this can lead to numerical oscillations near strong discontinuities (Figures 5(d) and (p)).

At finer resolutions (1200^2 and 2400^2), the IMWENO-P scheme emerges as particularly effective, achieving the lowest artificial heating among all schemes. The superior performance can be attributed to its symmetry-preserving property and stronger satisfaction of the sufficient condition as in

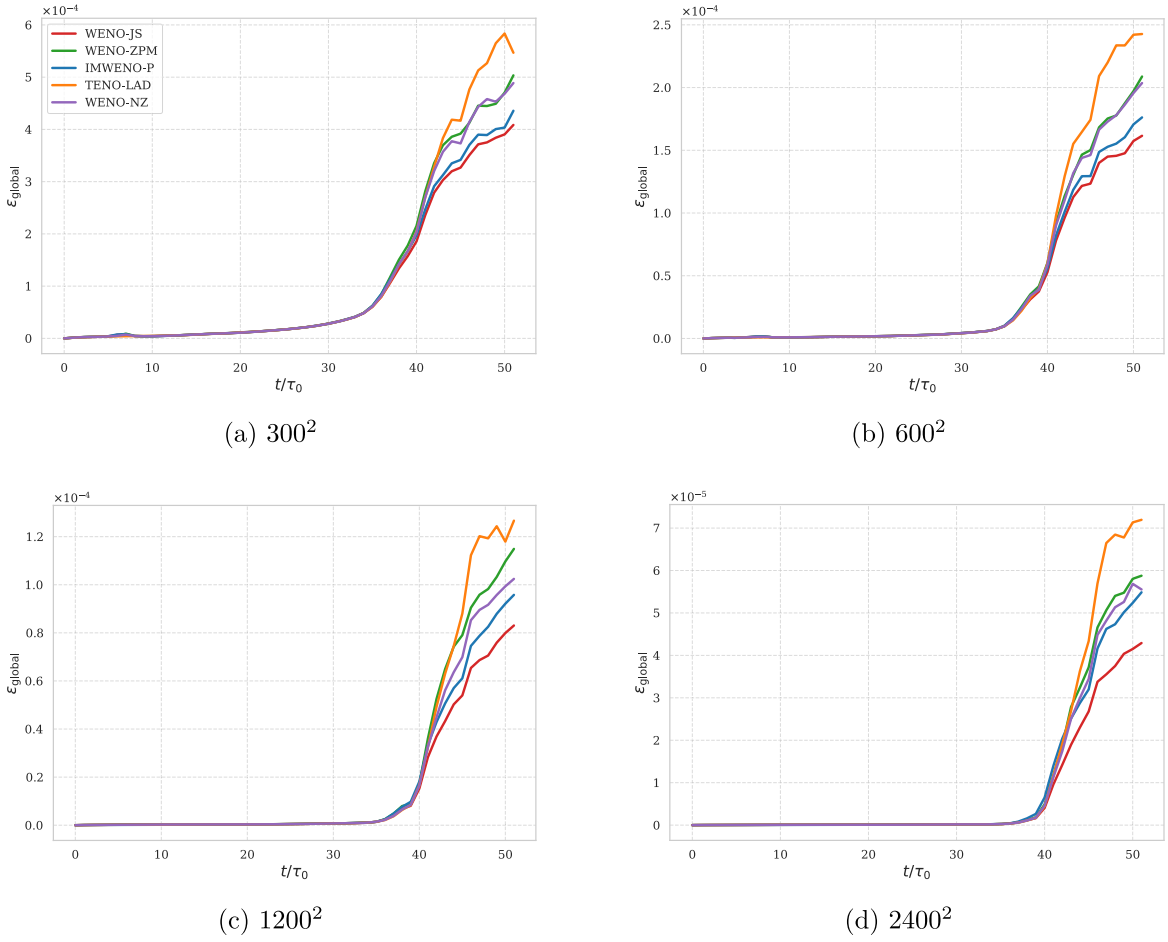


Figure 7. Time evolution of ϵ_{global} for different numerical schemes at various mesh resolutions. Each panel corresponds to a different mesh size, and all schemes are compared within each panel. The error metric accounts for both the magnitude of divergence errors and the magnetic field strength across the entire computational domain. Higher values indicate greater deviation from the $\nabla \cdot \mathbf{B} = 0$ condition.

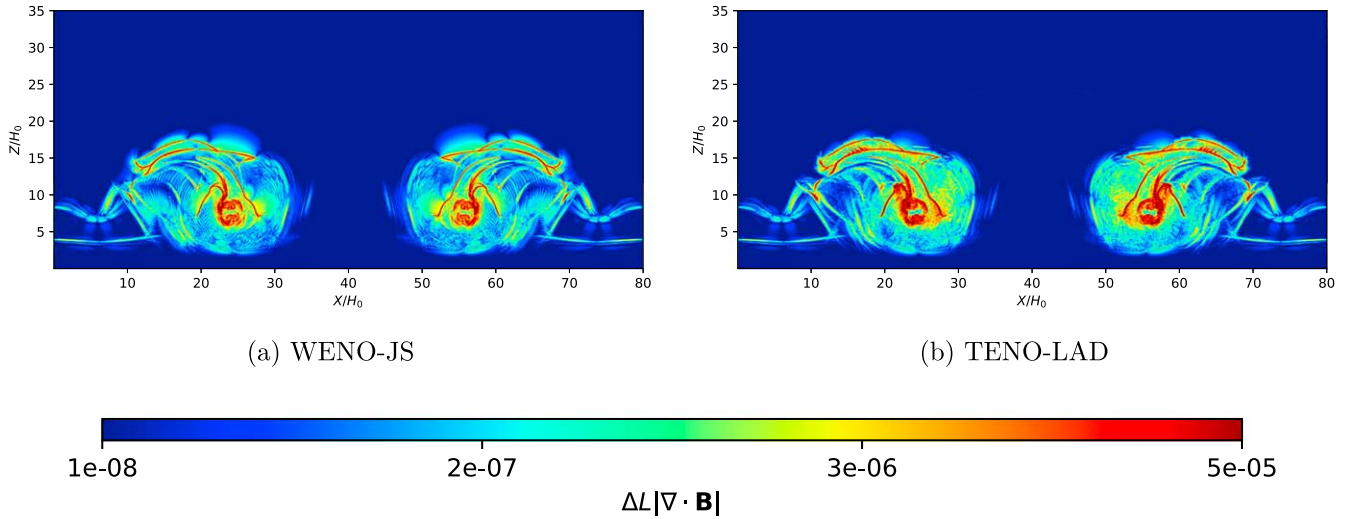


Figure 8. Spatial distribution of $\Delta L \|\nabla \cdot \mathbf{B}\|$ at $t/\tau_0 = 51$ for WENO-JS and TENO-LAD schemes at the finest resolution (2400^2). The color scale indicates the local magnitude of divergence errors. Both schemes exhibit increased divergence errors near current sheets, where rapid changes in field direction occur.

Equation (34), where the resolution-dependent LSIs in Equation (31) play a crucial role in the scheme's effectiveness. This enables the scheme to accurately distinguish between genuine discontinuities and high-wavenumber physical fluctuations, thereby applying appropriate levels of numerical diffusion only where necessary. As a result, the scheme can

maintain sharper resolution of discontinuities while preserving the physical features in smooth regions, leading to minimal spurious thermal energy generation.

The significance of these numerical characteristics is particularly evident given our use of a reduced adiabatic index ($\gamma = 1.05$) to approximate quasi-isothermal conditions. Under

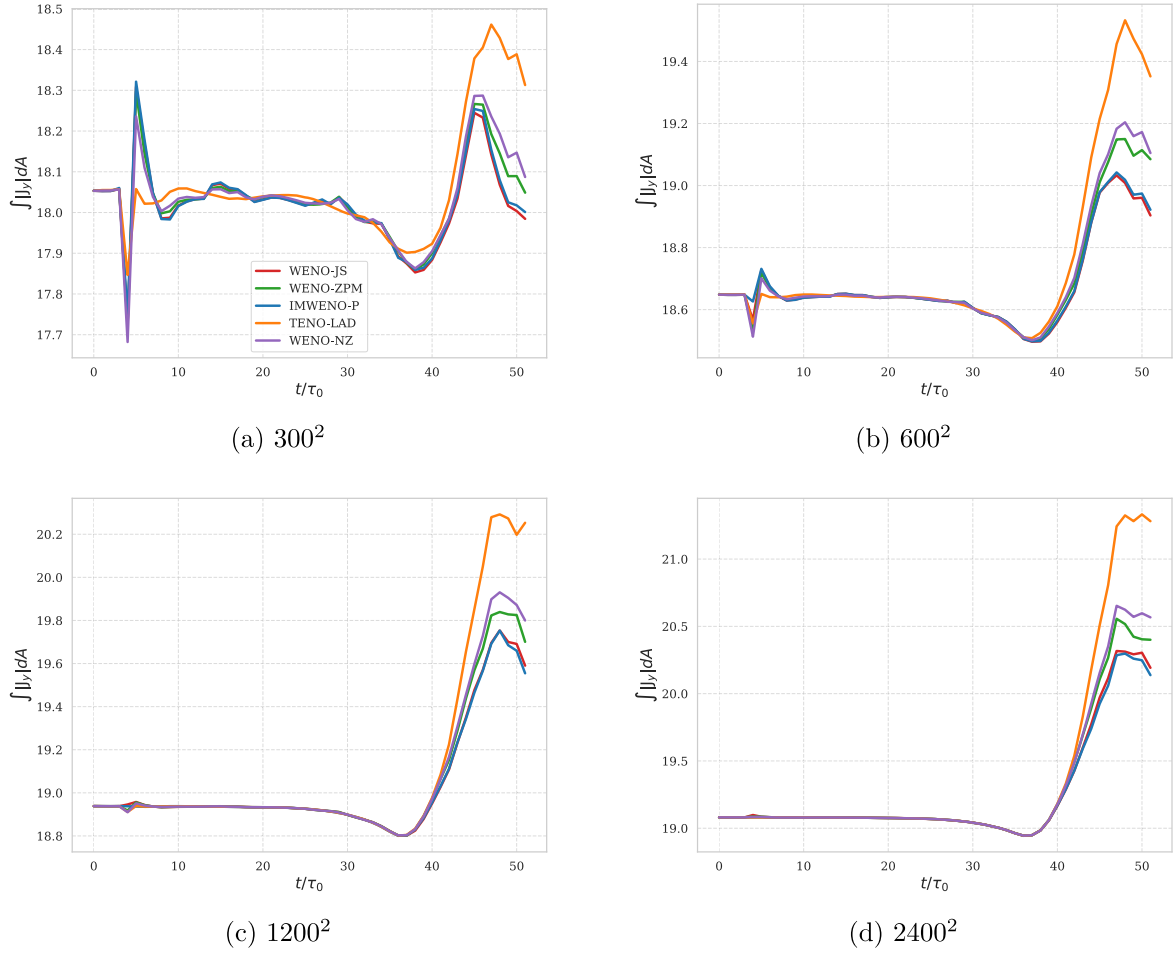


Figure 9. Time evolution of integrated current density for different numerical schemes at various mesh resolutions. Each panel corresponds to a different mesh size, and all schemes are compared within each panel. The differences between schemes reflect their varying abilities to resolve sharp magnetic features.

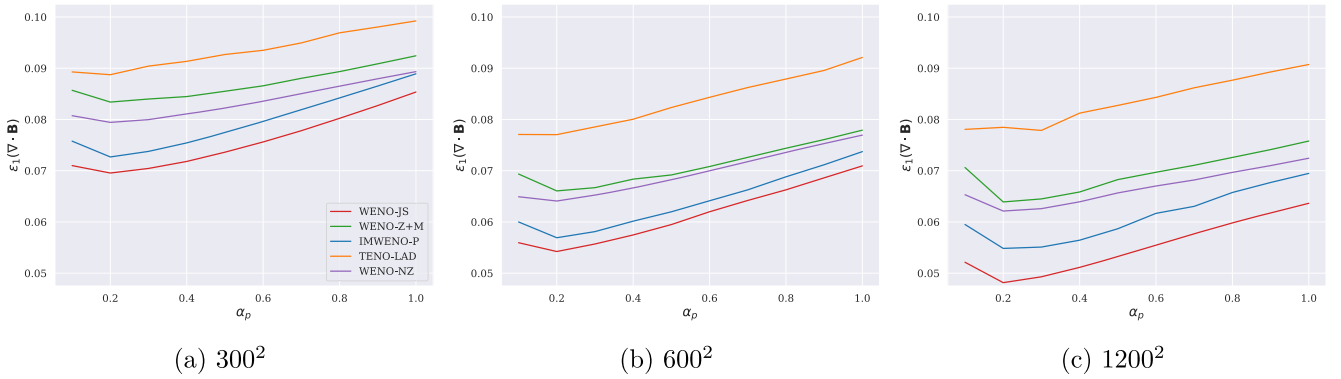


Figure 10. Analysis of magnetic field divergence errors ϵ_1 as a function of the damping parameter α_p at different mesh resolutions (300^2 , 600^2 , and 1200^2). WENO-family schemes show consistent optimal performance at $\alpha_p \approx 0.2$ across all resolutions, while TENO-LAD exhibits resolution-dependent behavior. Lower values in the plots indicate better divergence control.

these conditions, thermal energy should exhibit minimal growth during compressional processes, making any substantial thermal energy growth indicative of numerical artifacts rather than physical heating.

From a physical perspective, an analysis of the finest resolution cases (Figure 11, rightmost column) reveals the fundamental energy conversion mechanisms driving flux emergence. While magnetic buoyancy initiates the process, the primary driver of the evolution is the release of gravitational potential energy from magnetically supported mass distributions. Magnetic forces first

convert magnetic energy into kinetic energy, driving upward plasma flows. This kinetic energy then transforms to gravitational potential energy during ascent, before gravitational forces reconverge it to kinetic energy during descent. Throughout this process, plasma compression efficiently converts kinetic energy to thermal energy. This sequence aligns with theoretical predictions (T. C. Mouschovias 1974; R. Matsumoto et al. 1988, 1990). Several key features characterize this energy conversion process. The kinetic energy attains a minor component, as most energy is ultimately used in compressing the plasma. A strong temporal

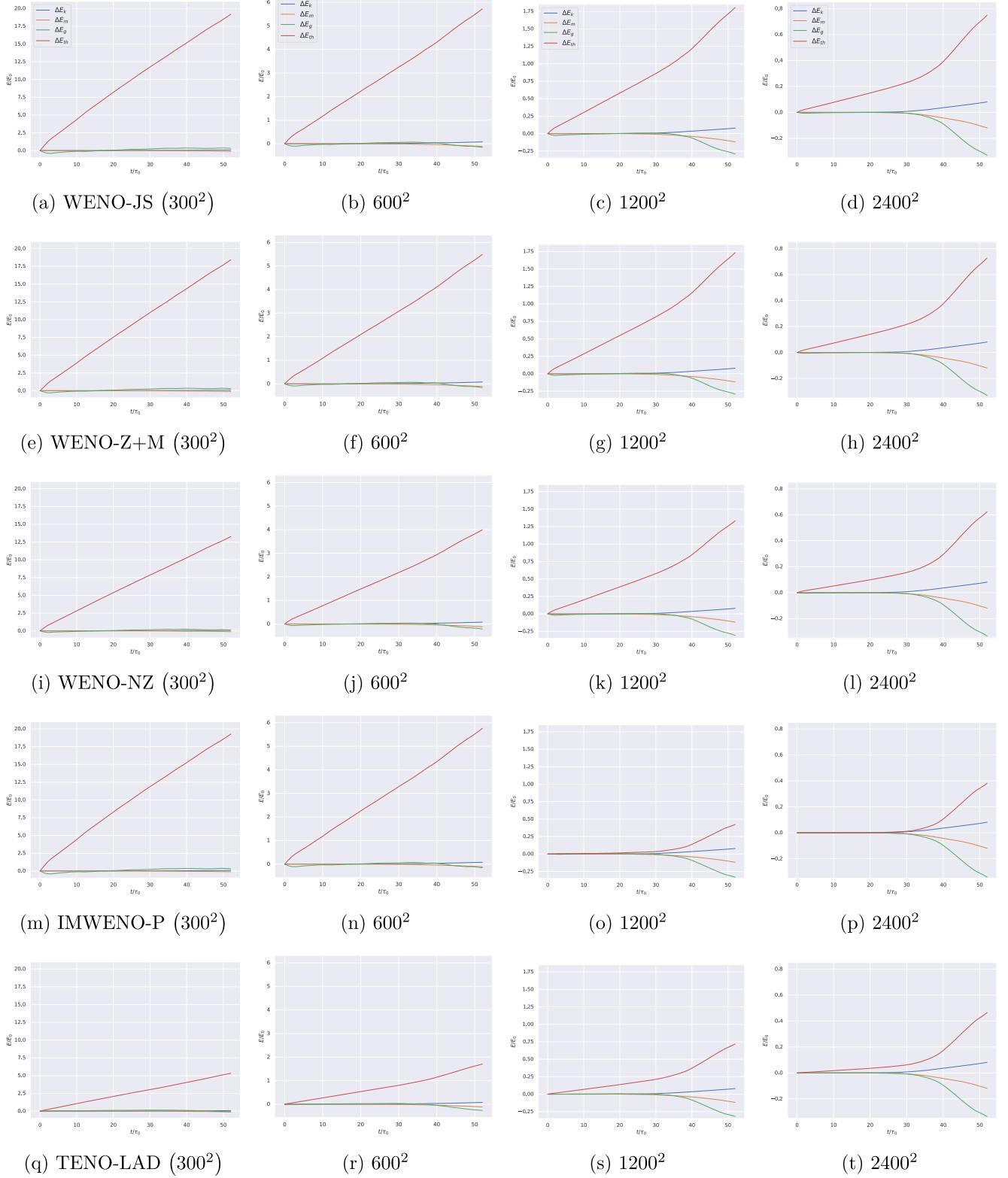


Figure 11. Evolution of system energetics ΔE_k , ΔE_m , ΔE_g , and ΔE_{th} across multiple resolutions (left to right: 300^2 to 2400^2). TENO-LAD exhibits superior performance with minimal artificial heating at coarse resolutions (300^2 and 600^2), while IMWENO-P achieves better energy conservation at finer resolutions (1200^2 and 2400^2).

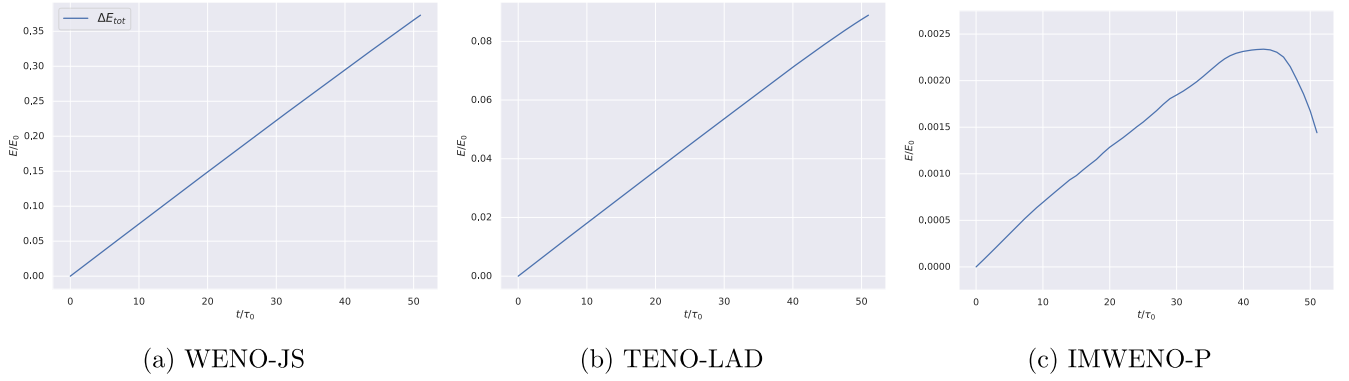


Figure 12. Comparison of the total energy evolution (ΔE_{tot}) at 2400^2 resolution for three different numerical schemes (WENO-JS, TENO-LAD, and IMWENO-P). While open boundary conditions predict a natural decrease in total energy as the magnetic loop rises, only the IMWENO-P scheme reproduces this expected behavior, highlighting its superior physical fidelity.

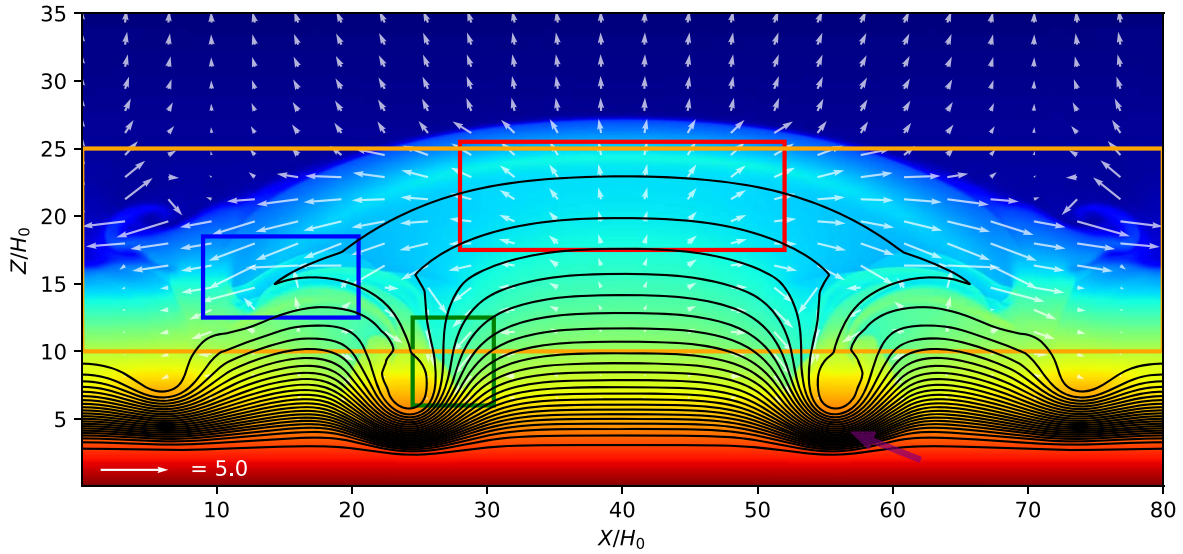


Figure 13. Computational domain showing ROIs R1–R4 (colored in red, blue, green, and orange, respectively) for mass and energy conservation analysis. The purple arrow indicates the location of a magnetic field valley.

correlation between kinetic and magnetic energies reflects the fundamental MHD coupling in our system. Notably, the magnetic energy shows minimal reduction despite significant spatial expansion of the field lines. This behavior stems from enhanced magnetic compression within valleys,⁹ which compensates for expansion-induced energy reduction, and the large coronal pressure scale height that restricts magnetic field expansion. As ambient gas pressure gradually decreases with height, the resulting pressure gradients continuously decelerate expanding loops, preventing free magnetic field expansion (K. Shibata et al. 1989b). In this context, the quasi-isothermal condition ($\gamma = 1.05$) helps distinguish between physical and numerical heating. While local shock compression naturally converts kinetic to thermal energy, significant global thermal energy increases beyond the gravitational energy release likely indicate numerical dissipation rather than physical processes. This interpretation is further supported by the total energy evolution (ΔE_{tot}) shown in Figure 12. Given the open boundary at the top of the computational domain, the total

energy should naturally decrease as the magnetic loop approaches this boundary. Among all tested schemes, only IMWENO-P reproduces this expected physical behavior, while other schemes exhibit unphysical continuous energy increases. The minimal amplitude of total energy variations in the IMWENO-P result provides additional evidence of its superior physical fidelity.

To quantify conservation properties, we examine mass and energy evolution within specific regions of interest (ROIs) selected for their physical significance (e.g., steep gradients or shock interactions). The analysis involves calculating the time variation of the total quantities (mass and energy) within the ROIs and the net fluxes across their boundaries. The conservation principle requires

$$\begin{aligned} \frac{dQ}{dt} &= -(\text{Flux}_{\text{out}} - \text{Flux}_{\text{in}}) \\ &= \text{Flux}_{\text{in}} - \text{Flux}_{\text{out}} = \text{Net Flux}, \end{aligned} \quad (49)$$

where Q represents conserved quantities (mass or energy). The ROIs are defined by their spatial boundaries illustrated in Figure 13. In our FV framework, the boundaries align with cell interfaces, where fluxes are naturally computed. The total mass

⁹ Magnetic field valleys refer to the U-shaped regions that form between rising magnetic loops, where magnetic field lines are compressed, and plasma accumulates (see Figure 13).

Table 2

Comparison of Mass and Energy Conservation Errors for Different Numerical Schemes across Four ROIs

Scheme	Type	$\epsilon_{R1}^{\text{avg}}$	$\epsilon_{R2}^{\text{avg}}$	$\epsilon_{R3}^{\text{avg}}$	$\epsilon_{R4}^{\text{avg}}$
WENO-JS	Mass	1.01e-06	4.13e-06	1.34e-05	1.29e-05
	Energy	6.63e-05	1.50e-04	1.18e-03	8.05e-04
WENO-Z+M	Mass	1.01e-06	4.10e-06	1.44e-05	1.31e-05
	Energy	6.63e-05	1.49e-04	1.19e-03	7.92e-04
WENO-NZ	Mass	1.01e-06	4.06e-06	1.30e-05	1.27e-05
	Energy	6.63e-05	1.48e-04	1.20e-03	7.58e-04
IMWENO-P	Mass	1.01e-06	3.89e-06	1.15e-05	9.95e-06
	Energy	6.60e-05	1.42e-04	1.17e-03	7.10e-04
TENOLAD	Mass	1.01e-06	3.98e-06	1.57e-05	9.76e-06
	Energy	6.59e-05	1.45e-04	1.21e-03	7.22e-04

Note. R1–R4 represent ROIs in Figure 13. The errors are computed using Equation (55). Bold values indicate the minimum error achieved in each ROI.

$M_{\text{ROI}}(t)$ and total energy $E_{\text{ROI}}(t)$ within an ROI at time t are computed by integrating the mass density ρ and total energy density e over the ROI:

$$M_{\text{ROI}}(t) = \iint_{\text{ROI}} \rho(x, z, t) dA, \quad (50)$$

$$E_{\text{ROI}}(t) = \iint_{\text{ROI}} e(x, z, t) dA. \quad (51)$$

The fluxes (F_m and F_e) through the ROI boundaries are defined as

$$F_m = \int_{\text{boundary}} \rho (\mathbf{V} \cdot \mathbf{n}) dl, \quad (52)$$

$$F_e = \int_{\text{boundary}} \left[\left(e + p + \frac{B^2}{2} \right) \mathbf{V} - (\mathbf{V} \cdot \mathbf{B}) \mathbf{B} \right] \cdot \mathbf{n} dl, \quad (53)$$

where $\mathbf{n}$ is the outward normal unit vector to the boundary, and dl is the differential length element along the boundary. The net flux into the ROI is expressed as

$$\text{Net Flux} = F_q^{\text{left}} + F_q^{\text{right}} + F_q^{\text{bottom}} + F_q^{\text{top}} = F_q^{\text{net}}, \quad (54)$$

with q representing either mass or energy. Conservation errors are quantified by

$$\epsilon^{\text{avg}} = \frac{1}{T} \int_0^T \epsilon_{\text{ROI}}(t) dt \quad \text{with} \quad \epsilon_{\text{ROI}}(t) = \left| \frac{dQ_{\text{ROI}}}{dt} - F_q^{\text{net}}(t) \right|, \quad (55)$$

where T denotes the total simulation time. The results, presented in Table 2, indicate the performance of each numerical scheme in conserving mass and energy within the ROIs.

These findings emphasize that an accurate capture of energy evolution requires an appropriate numerical scheme selection and mesh resolution. While TENOLAD demonstrates the best performance in minimizing thermal energy generation at coarse resolutions, IMWENO-P outperforms all other schemes at finer resolutions (1200^2 and 2400^2). As demonstrated by E. Johnsen et al. (2010), sufficient mesh resolution is also crucial to resolve physical over numerical diffusion, particularly in regions with interspersed shocks and turbulence. The conservation analysis through ROIs (Table 2) further validates IMWENO-P's

superior performance, as it exhibits comparable and lower conservation errors across most regions compared to other schemes. The combination of an appropriate numerical scheme selection and adequate mesh resolution thus enables a reliable investigation of the complex energy conversion processes in magnetic flux emergence.

5. Conclusion

We have presented a comparative assessment of advanced HRSC reconstruction schemes for two-dimensional MHD simulations of magnetic flux emergence in a stratified solar atmosphere. Through the detailed analysis of WENO-family schemes (WENO-JS, WENO-Z+M, WENO-NZ, IMWENO-P) and a TENO-family scheme (TENOLAD), we demonstrate that variations in nonlinear weighting strategies, smoothness indicators, and stencil selection substantially impact the capture of subtle MHD processes, particularly in quasi-isothermal conditions with $\gamma = 1.05$.

While all schemes successfully reproduce the fundamental features of the Parker instability including buoyant magnetic flux rise, loop formation, and associated MHD shocks (fast, slow, and intermediate), their performance characteristics diverge significantly beyond these basic features. High-diffusion schemes like WENO-JS effectively control magnetic divergence errors and smooth out small-scale current structures, but introduce artificial global expansion and spurious thermal energy increases. In contrast, low-diffusion approaches such as TENOLAD better preserve finer features and minimize spurious heating at coarse resolutions, while facing challenges with magnetic divergence control and numerical oscillations near strong discontinuities. Among the tested methods, the IMWENO-P stands out at fine resolutions, largely due to its LSIs that possess both resolution-dependent and symmetry-preserving properties. It effectively minimizes spurious thermal energy generation while maintaining stable compression patterns throughout the domain. Most importantly, the observed thermal energy increases align with physical expectations, originating primarily from gravitational potential energy release rather than artificial compression effects, which is particularly crucial in our quasi-isothermal setup. The resulting energy redistribution patterns closely match theoretical predictions for solar flux emergence, where the dynamics are dominated by gravitational potential energy release and local shock dissipation. This physical consistency enables more reliable shock identification and flow diagnostics. In contrast, other schemes often exhibit unphysical behavior, manifesting as contradictory combinations of negative compressional work integrals (indicating net expansion) and spurious thermal energy increases.

These findings underscore the importance of selecting appropriate numerical schemes, resolution levels, and model parameters for accurately capturing the complex dynamics of solar MHD phenomena. Our comprehensive evaluation across multiple physical metrics, rather than relying on a single measure of accuracy, reveals that subtle differences in numerical algorithms can significantly impact the evolution of global flow patterns, shock formation, magnetic field structures, and energy pathways. Parameter studies, such as fine-tuning damping parameters (α_p), may further enhance the robustness and accuracy of these approaches.

Extending this analysis to three-dimensional configurations, incorporating additional physical effects (e.g., partial ionization or radiative cooling), and employing adaptive mesh refinement strategies, and advanced HLLD Riemann solvers

(T. Matsumoto et al. 2019; Y. Matsumoto et al. 2019; T. Minoshima et al. 2020; T. Minoshima & T. Miyoshi 2021) represent directions for future research. Through the continued refinement of HRSC schemes and their implementation in complex MHD environments, it becomes increasingly feasible to capture the nuanced dynamics of flux emergence, advance our understanding of solar atmospheric behavior, and improve predictive capabilities for solar eruptive events.

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indebted to the developers and maintainers of the open-source software packages that made this research possible.

Software: OpenMP (L. Dagum & R. Menon 1998), Intel® MPI (<https://www.intel.com/content/www/us/en/docs/mpi-library/developer-reference-linux/2021-11/overview.html>), NumPy (C. R. Harris et al. 2020), SciPy (P. Virtanen et al. 2020), Matplotlib (J. D. Hunter 2007), Seaborn (M. L. Waskom 2021).

Appendix A Temporal Evolution of Parker Instability

The temporal evolution of the Parker instability from $t = 0$ to $t = 47\tau_0$ is illustrated in Figure 14, using the IMWENO-P scheme at 2400^2 resolution. These snapshots show the progressive development of the instability before reaching the most dynamically active state at $t = 51\tau_0$ (see Figure 2).

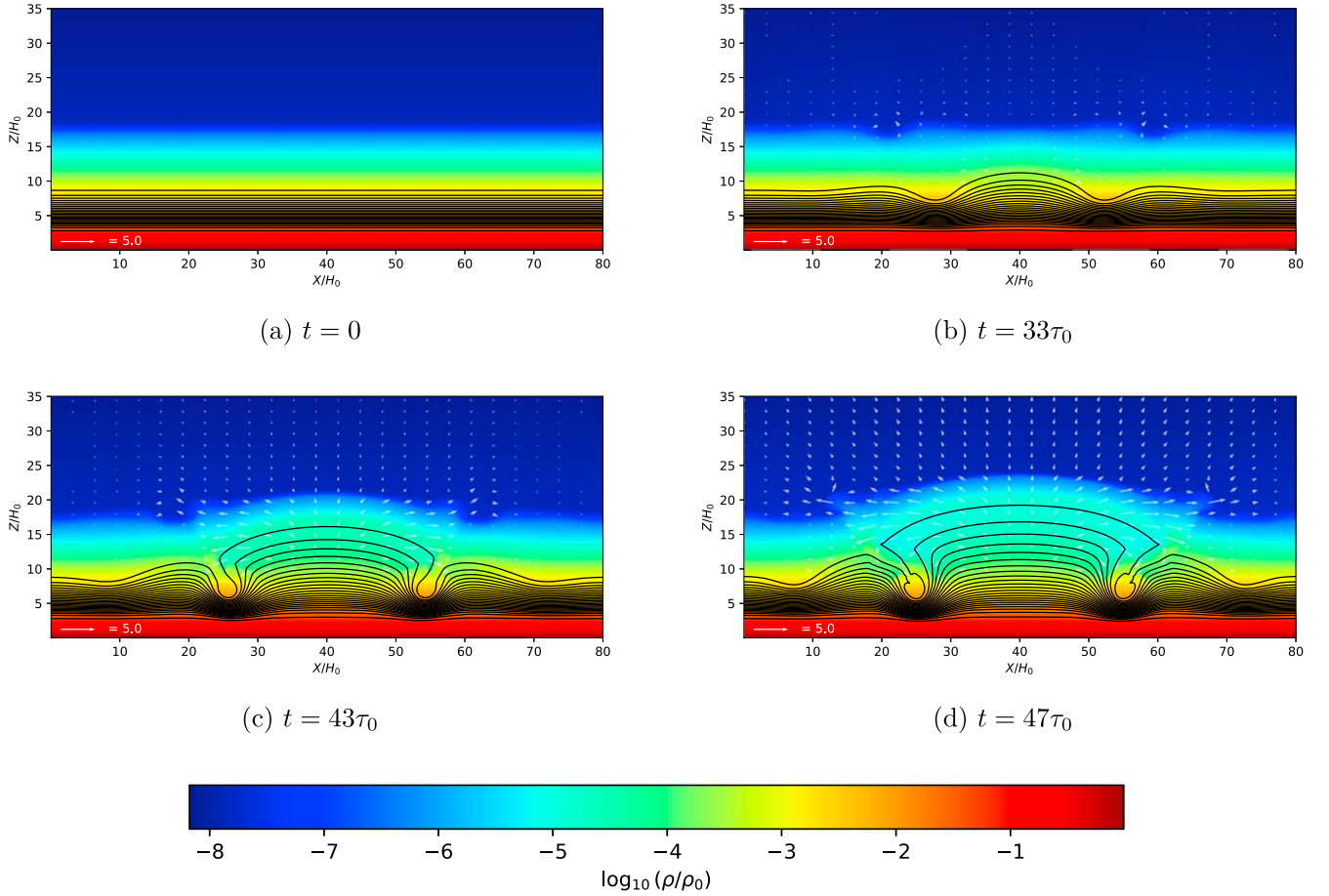


Figure 14. Time evolution of magnetic field lines (black curves), velocity vectors (white arrows), and density distribution (color) at $t = 0, 33\tau_0, 43\tau_0, 47\tau_0$ using IMWENO-P at 2400^2 resolution.

Appendix B

Resolution Effects on Self-similar Expansion Parameters

This appendix presents a resolution-dependent analysis of key physical quantities under the self-similar expansion law

(see Section 4.1). Results for all numerical schemes are shown at resolutions of 600^2 (Figure 15), 1200^2 (Figure 16), and 2400^2 (Figure 17) to examine convergence and accuracy.

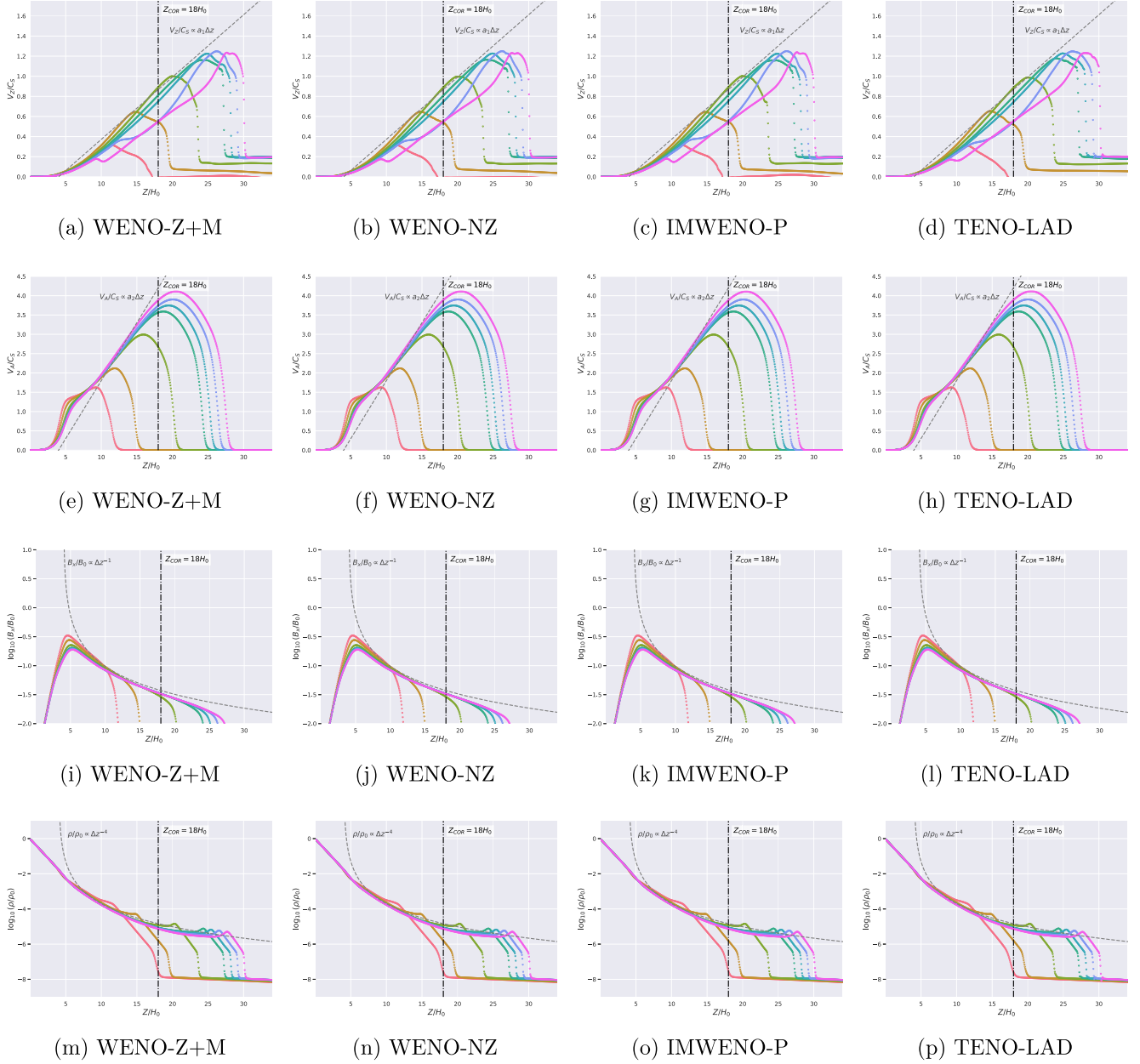


Figure 15. Comparison of key physical quantities at 600^2 resolution using different numerical schemes. All quantities are evaluated at $x/H_0 = (X_{\max}/2)/H_0$, corresponding to the midpoint of the rising loop. The TENO-LAD scheme still shows oscillations in the rising velocity (d) and density profiles (p), indicating underresolved gradients. Other remarks are the same as in Figure 5.

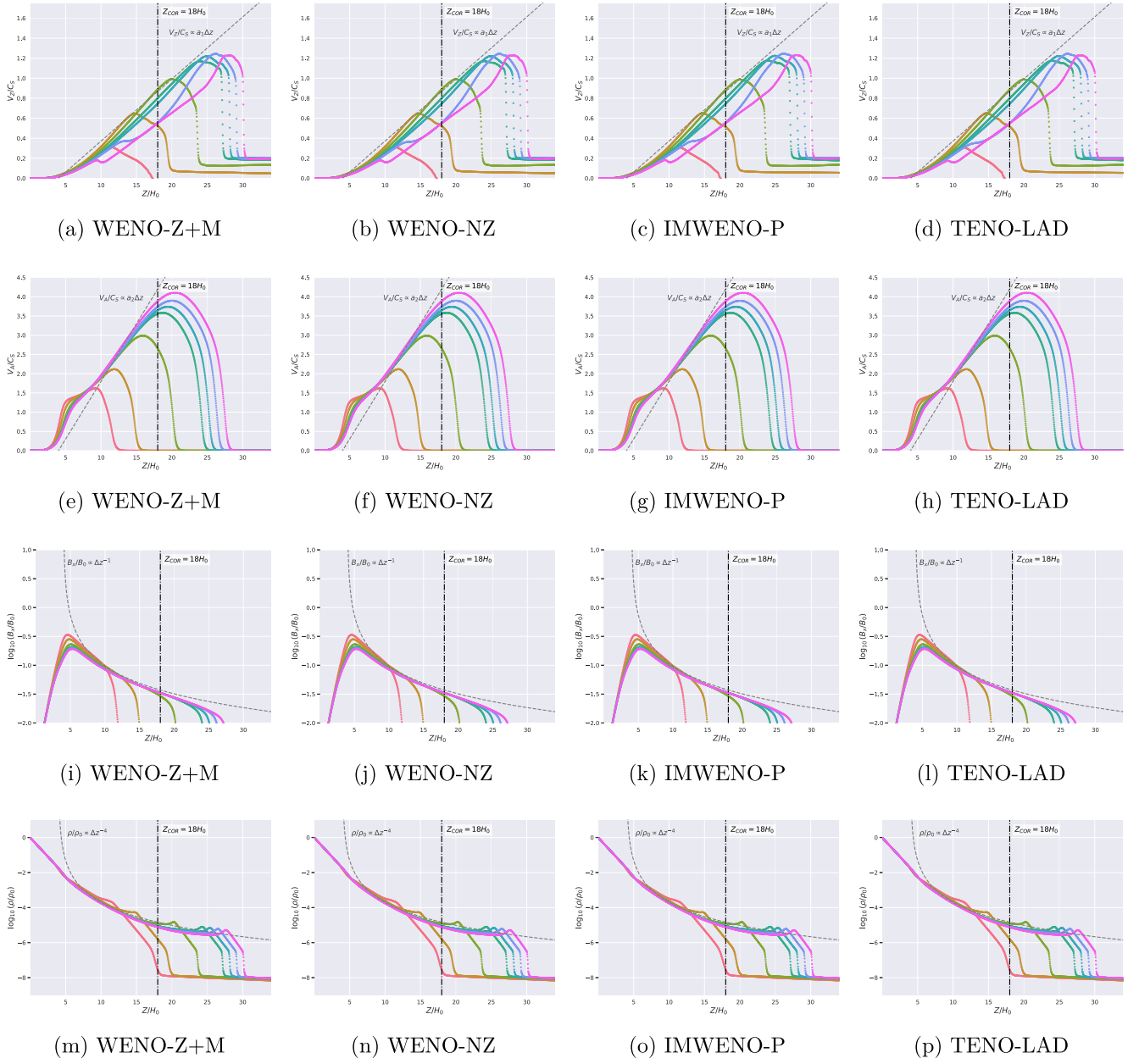


Figure 16. Comparison of key physical quantities at 1200^2 resolution using different numerical schemes. All quantities are evaluated at $x/H_0 = (X_{\max}/2)/H_0$, corresponding to the midpoint of the rising loop. Other remarks are the same as in Figure 5.

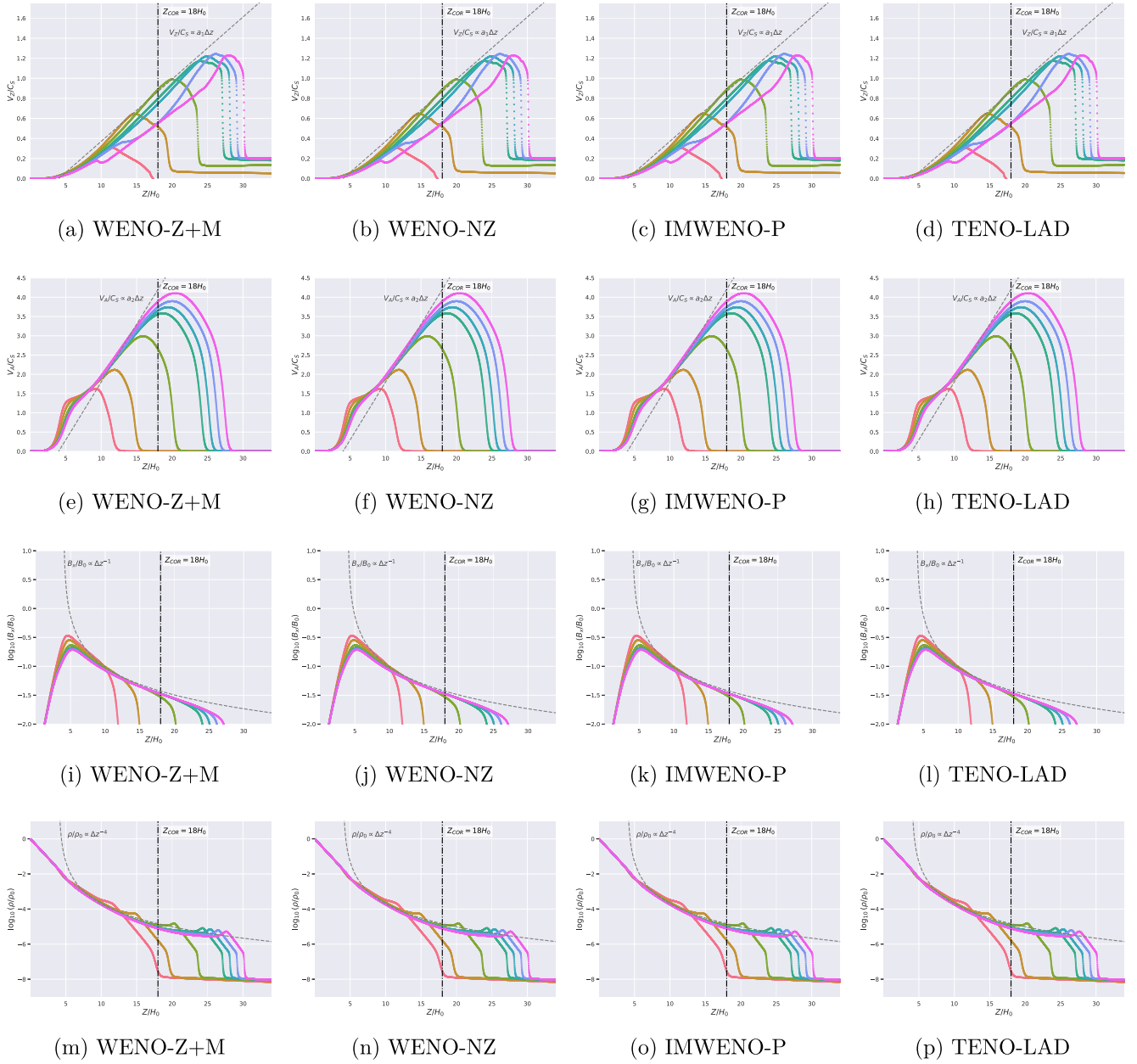


Figure 17. Comparison of key physical quantities at 2400^2 resolution using different numerical schemes. All quantities are evaluated at $x/H_0 = (x_{\text{max}}/2)/H_0$, corresponding to the midpoint of the rising loop. Other remarks are the same as in Figure 5.

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